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FACULTY OF COMPUTER SCIENCE AND ENGINEERING**



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SLURM AGENT SPECILIST

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Declaration of Authenticity

We declare that we solely conducted this specialized project, under the supervision of Assoc.Prof. Thoại Nam at the Faculty of Computer Science and Engineering, Vietnam National University - Ho Chi Minh City University of Technology.

We have taken care to properly acknowledge and document all external sources and references used in the project.

If there is any instance of plagiarism, we are ready to accept the consequences. Ho Chi Minh City University of Technology - Vietnam National University HCMC will not be held responsible for any copyright violations that may have occurred during my research.

Ho Chi Minh City, December 2024

Authors,

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Abstract

High-Performance Computing (HPC) clusters have become essential infrastructure for scientific research, engineering simulations, and large-scale data processing. The Slurm workload manager, deployed on the majority of the world’s top supercomputers, provides sophisticated resource management capabilities but presents significant usability challenges due to its complex command-line interface. This thesis addresses these challenges by designing and implementing an intelligent agent system that provides a natural language interface for Slurm cluster management.

The proposed Slurm Agent leverages Large Language Models (LLMs) combined with the Model Context Protocol (MCP) to bridge the gap between user intent and system operations. The system implements a multi-agent architecture where specialized sub-agents handle different aspects of cluster interaction: an Analysis Sub-Agent for read-only queries, an Action Sub-Agent for job management, plus auxiliary tools for data visualization and documentation retrieval. Through MCP-based tool integration, the agent executes Slurm commands, interprets results, and presents information in accessible formats.

Key contributions of this research include: (1) a natural language interface enabling users to query cluster state without command syntax knowledge, (2) an MCP server implementation extending the protocol ecosystem to the HPC domain, (3) a multi-agent architecture with handoff mechanisms for specialized task handling, (4) integrated visualization capabilities for cluster metrics, and (5) real-time streaming response delivery using Server-Sent Events.

Experimental evaluation on a single-node Slurm cluster demonstrates high accuracy (96.9%) in response correctness, responsive streaming interaction with time-to-first-token under 2 seconds, and effective handling of diverse query types including status

queries, visualization requests, and multi-turn conversations.

The implementation provides a foundation for continued research in AI-assisted HPC operations, demonstrating the viability of LLM-based interfaces for complex system management while identifying opportunities for future enhancement including write operation support, local model deployment, and multi-cluster management.

Keywords: High-Performance Computing, Slurm, Large Language Models, Model Context Protocol, Multi-Agent Systems, Natural Language Interface

List of Abbreviations

Abbreviation	Description
API	Application Programming Interface
CLI	Command-Line Interface
CoT	Chain-of-Thought
CPU	Central Processing Unit
GPU	Graphics Processing Unit
HPC	High-Performance Computing
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
JSON-RPC	JavaScript Object Notation Remote Procedure Call
LLM	Large Language Model
MCP	Model Context Protocol
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
ReAct	Reasoning and Acting
REST	Representational State Transfer
RLHF	Reinforcement Learning from Human Feedback
SDK	Software Development Kit
Slurm	Simple Linux Utility for Resource Management
SSE	Server-Sent Events
TLS	Transport Layer Security
UI	User Interface

Abbreviation	Description
URI	Uniform Resource Identifier

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Chapter 1

Introduction

1.1 Motivation

High-Performance Computing (HPC) has become an essential infrastructure in modern scientific research, engineering simulations, and large-scale data processing. HPC clusters enable researchers to tackle computationally intensive problems that would be infeasible on conventional computing systems. From climate modeling and genomics to machine learning and financial simulations, HPC systems provide the computational power necessary to drive innovation across multiple domains.

The Simple Linux Utility for Resource Management (Slurm) has emerged as the dominant workload manager for HPC clusters, managing resources in approximately 60% of the world's top supercomputers [23, 15]. Slurm provides sophisticated mechanisms for job scheduling, resource allocation, and cluster monitoring, enabling efficient utilization of computational resources across thousands of nodes. The system's flexibility and scalability have made it the preferred choice for both academic institutions and commercial organizations operating large-scale computing facilities.

Despite its power and flexibility, Slurm presents significant usability challenges that create barriers for both novice and experienced users [14]. The system exposes its functionality through a complex command-line interface (CLI) comprising numerous commands, each with extensive sets of options and parameters. Users must master commands such as `squeue`, `sinfo`, `sbatch`, `scontrol`, and `sacct`, along with their associated flags and output formats. This complexity is compounded by the need to write job submis-

sion scripts in Bash, which requires additional technical knowledge beyond the domain expertise of many researchers.

The learning curve associated with Slurm administration and usage creates several operational challenges. System administrators spend considerable time training new users, while researchers face delays in their work as they navigate the intricacies of job submission and monitoring. Common tasks such as checking job status, understanding resource availability, or debugging failed jobs often require consulting extensive documentation or seeking expert assistance.

Recent advances in Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language understanding and generation [4, 10]. These models can interpret complex user queries, reason about multi-step tasks, and generate appropriate responses or actions. The emergence of LLM-based agents—systems that combine language models with tool-calling capabilities—has opened new possibilities for creating intelligent interfaces to complex systems [21, 16].

The Model Context Protocol (MCP), developed by Anthropic, provides a standardized approach for connecting LLMs to external tools and data sources [1]. MCP enables the creation of modular, reusable integrations that can expose system functionality to AI agents in a secure and controlled manner. This architectural pattern separates the concerns of tool implementation from agent logic, facilitating the development of sophisticated AI-assisted applications.

This research is motivated by the opportunity to bridge the gap between Slurm’s powerful capabilities and user accessibility through an intelligent conversational interface. By leveraging LLM agents connected to Slurm through MCP, we can create a system that allows users to manage cluster resources using natural language queries, eliminating the need to memorize complex command syntax and reducing the cognitive load associated with HPC operations.

1.2 Scope and Goals

1.2.1 Project Goals

The primary objective of this research is to design and implement an intelligent agent system that provides a natural language interface for Slurm cluster management. This system aims to democratize access to HPC resources by enabling users to interact with Slurm using conversational queries rather than complex command-line syntax. The following goals guide the development of this solution:

Goal 1: Natural Language Interface for Slurm Operations. The system shall enable users to perform common Slurm operations through natural language queries. Users should be able to check job status, view cluster resources, submit jobs, and monitor queues without explicit knowledge of Slurm command syntax. The agent must interpret user intent accurately and translate queries into appropriate Slurm commands.

Goal 2: Multi-Agent Architecture with Specialized Capabilities. The system shall implement a multi-agent architecture where specialized agents handle different aspects of cluster management. This includes agents for command execution, data visualization, and web search. The handoff mechanism between agents should be seamless and transparent to the user.

Goal 3: Standardized Tool Integration via MCP. The system shall utilize the Model Context Protocol to provide standardized tool interfaces between the LLM agents and Slurm infrastructure. The MCP server should expose Slurm functionality as discrete tools that can be invoked by the agent, ensuring clean separation between the AI layer and system operations.

Goal 4: Visualization of Cluster Metrics. The system shall provide visualization capabilities for cluster data, enabling users to request charts and graphs showing resource utilization, job distributions, and other metrics. Visualizations should be generated dynamically based on user queries and current cluster state.

Goal 5: Real-Time Streaming Responses. The system shall support real-time streaming of agent responses using Server-Sent Events (SSE), providing users with immediate feedback during long-running operations. This improves user experience by eliminating

the perception of system unresponsiveness.

Goal 6: Web-Based User Interface. The system shall provide an intuitive web-based interface that supports markdown rendering, code highlighting, and inline visualization display. The interface should be accessible and usable without specialized client software.

1.2.2 Project Scope

The scope of this project encompasses the design, implementation, and evaluation of the Slurm Agent system. The following boundaries define what is included and excluded from the project:

In Scope:

- Development of an MCP server exposing Slurm functionality as tools
- Implementation of a multi-agent system using the OpenAI Agents SDK
- Design and implementation of a web-based frontend interface
- Integration of chart visualization capabilities using matplotlib
- Implementation of web search functionality for documentation assistance
- Real-time response streaming using SSE
- Testing on a single-node Slurm cluster configuration

This project focuses on demonstrating the feasibility and effectiveness of using LLM agents with MCP for HPC cluster management, providing a foundation for future research and development in this domain.

1.3 Challenges and Solutions

The development of an LLM-based agent for Slurm cluster management presents several technical and architectural challenges. This section identifies these challenges and outlines the approaches adopted to address them.

1.3.1 Challenge 1: Bridging Natural Language to System Commands

Challenge Description: Slurm operations require precise command syntax with specific parameters and flags. Natural language queries are inherently ambiguous and may not provide all necessary information for command execution. The system must accurately interpret user intent and request clarification when required, while avoiding incorrect command execution that could impact cluster operations.

Solution Approach: We employ a ReAct (Reasoning and Acting) architecture [22] where the main agent follows a Think $\rightarrow$ Act $\rightarrow$ Observe loop. The MCP server exposes Slurm commands as tools with well-defined Pydantic parameter schemas, enabling the LLM to understand required and optional parameters. The agent reasons about each query before acting, and can engage in multi-turn dialogue using SQLite-backed conversation history to gather missing information from users.

1.3.2 Challenge 2: Managing Complex Multi-Step Operations

Challenge Description: Many cluster management tasks require multiple sequential operations with dependencies between steps. For example, analyzing job performance may require first retrieving job information, then querying accounting data, and finally generating a visualization. Managing these workflows while maintaining coherent conversation context presents significant complexity.

Solution Approach: We implement a multi-agent architecture where the main ReAct agent orchestrates specialized sub-agents invoked as tools. An Analysis Sub-Agent handles read-only queries (cluster status, job information, web search), while an Action Sub-Agent manages job operations. Sub-agents return results to the main agent, which maintains overall context and coordinates execution. A `ChartFilteredSession` wrapper strips Mermaid chart code from conversation history before sending to the LLM, reducing token consumption since visualization code is verbose but not needed for reasoning.

1.3.3 Challenge 3: Real-Time Response Delivery

Challenge Description: LLM inference and Slurm command execution can introduce noticeable latency, potentially leading to poor user experience. Users may perceive the

system as unresponsive during processing, particularly for complex queries that require multiple tool invocations or lengthy computations.

Solution Approach: We implement Server-Sent Events (SSE) for real-time streaming of partial responses in OpenWebUI-compatible format. The backend generates SSE chunks containing incremental content, and the frontend displays updates as the agent generates responses. Chart artifacts are streamed separately and appended at stream end, enabling inline rendering within the conversation interface.

1.3.4 Challenge 4: Secure Tool Execution

Challenge Description: The system must execute Slurm commands on behalf of users, raising concerns about command injection, unauthorized operations, and potential system impact. Improper handling could allow malicious queries to execute harmful commands on the cluster.

Solution Approach: The system implements a layered security model. First, tools are categorized into safe read-only operations (squeue, sinfo, sacct) and dangerous write operations (scancel, sbatch, scontrol updates). Safe tools execute immediately, while dangerous tools are queued in a persistent Pending Actions Store requiring explicit user confirmation. The MCP server uses parameterized tool interfaces with Pydantic validation that prevent arbitrary command injection. Dynamic tool filters ensure sub-agents can only access their permitted tool sets.

1.3.5 Challenge 5: Visualization Generation and Delivery

Challenge Description: Generating and delivering data visualizations within a conversational interface presents unique challenges. The system must determine appropriate chart types based on data characteristics, generate visualizations dynamically, and deliver them through the SSE response stream in a format consumable by the web frontend.

Solution Approach: A dedicated chart generation tool handles visualization using Mermaid diagram syntax. The main agent calls the `generate_chart` tool via the MCP server, which returns Mermaid code for chart types including pie charts, bar charts, and flowcharts. Chart artifacts are stored in a context object during the run and transmitted through the SSE stream as Mermaid code blocks. The frontend renders these using

a Mermaid renderer, providing inline visualization within the conversation. Critically, a `ChartFilteredSession` wrapper makes the agent “blind” to chart artifacts—Mermaid blocks are stripped from conversation history before LLM inference, significantly reducing token consumption since chart code can be hundreds of tokens that provide no reasoning value.

1.4 Thesis Structure

This thesis is organized into seven chapters, each addressing specific aspects of the research and development of the Slurm Agent system. The structure is designed to provide a logical progression from theoretical foundations through implementation details to experimental evaluation.

Chapter 1: Introduction. This chapter presents the motivation for developing an LLM-based agent for Slurm cluster management. It establishes the scope and goals of the project, identifies key challenges encountered during development, and outlines the solutions adopted to address them.

Chapter 2: Theoretical Background. This chapter provides the foundational knowledge necessary to understand the research. It covers Slurm workload management, including architecture and key concepts; Large Language Models and their capabilities; and the Model Context Protocol for tool integration. These concepts form the technical basis upon which the solution is built.

Chapter 3: Related Tools and Works. This chapter surveys existing solutions and related research in the domains of HPC management interfaces, LLM-based agent frameworks, and MCP implementations. It analyzes the strengths and limitations of current approaches, positioning the contribution of this research within the broader landscape.

Chapter 4: Proposed Approaches. This chapter presents the design of the Slurm Agent system. It begins with an introduction to the application concept, followed by detailed requirements analysis. The chapter describes the overall system architecture and the multi-agent design that enables specialized handling of different task types.

Chapter 5: Implementation. This chapter details the implementation of each system component. It covers the MCP server development for exposing Slurm functionality,

the agent backend implementation using the OpenAI Agents SDK, and the frontend development including the SSE-based real-time update mechanism and visualization rendering.

Chapter 6: Experiments. This chapter describes the testing methodology and presents experimental results. It covers the testing approaches employed, the test environment configuration, and evaluation of the system’s capabilities across various usage scenarios.

Chapter 7: Conclusion. This chapter summarizes the research contributions and findings. It discusses limitations of the current implementation and identifies opportunities for future work and potential extensions of the system.

Chapter 2

Theoretical Background

2.1 Slurm Workload Manager

Slurm (Simple Linux Utility for Resource Management) is an open-source, highly scalable cluster management and job scheduling system designed for Linux clusters [23]. Originally developed at Lawrence Livermore National Laboratory, Slurm has become the predominant workload manager in the HPC industry, deployed on approximately 60% of the world's top 500 supercomputers. This section provides an overview of Slurm's architecture, core concepts, and operational mechanisms.

2.1.1 Architecture Overview

Slurm employs a centralized controller architecture with distributed compute node daemons. The system consists of three primary daemon components that work together to manage cluster resources and execute user workloads.

The **slurmctld** daemon serves as the central management controller, running on a dedicated management node. This daemon maintains the global state of the cluster, including information about nodes, partitions, jobs, and resource allocations. The controller is responsible for accepting job submissions, making scheduling decisions, and dispatching work to compute nodes. For high availability, Slurm supports backup controller configurations where a secondary **slurmctld** can assume control if the primary fails.

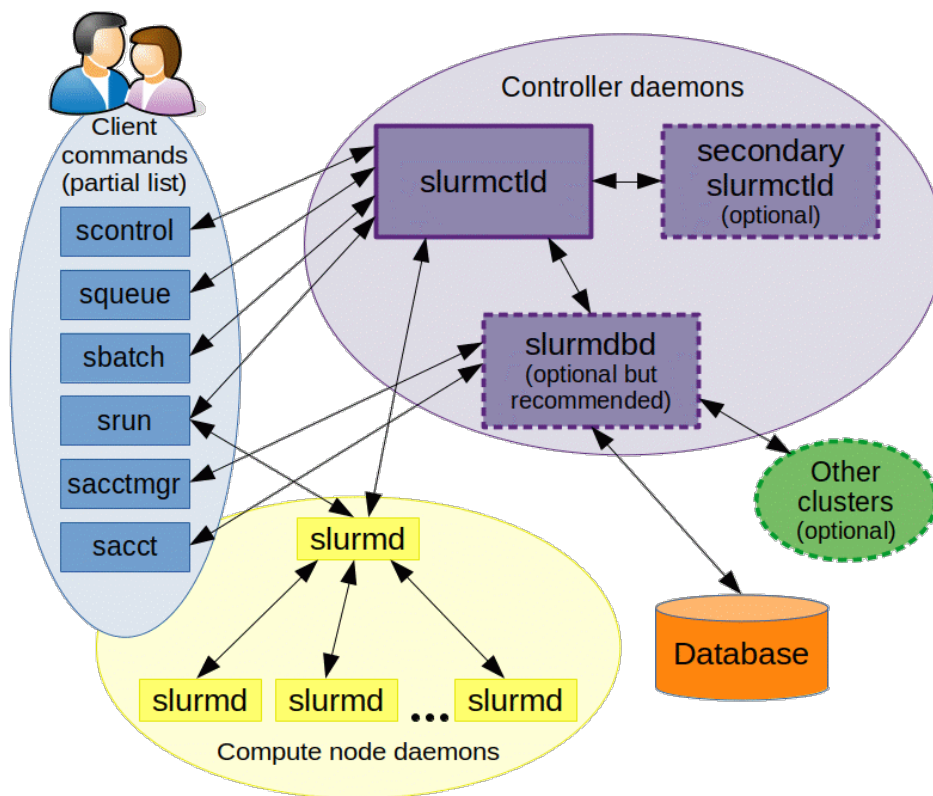


Figure 2.1: Slurm System Architecture

The **slurmd** daemon runs on each compute node in the cluster, acting as a remote execution agent. Each slurmd is responsible for launching, monitoring, and terminating jobs on its local node. The daemon reports node state and resource availability to the central controller and receives job execution requests. Slurmd ensures proper resource isolation for running jobs and handles signal propagation for job control.

The **slurmdbd** daemon (Slurm Database Daemon) provides an optional but commonly used accounting interface. This daemon manages connections to a backend database (typically MySQL or MariaDB) where job accounting records, resource usage statistics, and user associations are stored. The database daemon enables administrators to track resource consumption, generate usage reports, and implement fair-share scheduling policies.

2.1.2 Core Concepts

Understanding Slurm requires familiarity with several fundamental concepts that define how resources are organized and allocated.

Nodes represent individual compute servers in the cluster. Each node has defined resources including CPUs (or cores), memory, GPUs, and potentially other generic resources. Nodes can be in various states: idle (available), allocated (running jobs), down (unavailable), or drained (not accepting new work). Node features and constraints allow jobs to request specific hardware characteristics.

Partitions serve as logical groupings of nodes, functioning similarly to queues in other scheduling systems. Partitions define access controls, resource limits, and scheduling policies for different user groups or job types. A single node can belong to multiple partitions, and users submit jobs to specific partitions based on their requirements and permissions.

Jobs are resource allocation requests submitted by users. A job specification includes resource requirements (nodes, CPUs, memory, time limit), the program to execute, input/output file locations, and various optional parameters. Jobs progress through states including pending, running, completing, and completed (or failed/cancelled).

Steps are discrete execution units within a job. While a job represents the overall resource allocation, steps allow multiple parallel or sequential programs to execute within that allocation. Each step can use a subset of the job's allocated resources.

2.1.3 Command-Line Interface

Slurm provides a comprehensive suite of command-line utilities for cluster interaction:

- `sinfo`: Displays information about nodes and partitions, including their state, resources, and availability
- `squeue`: Shows the queue of pending and running jobs, with details about resource allocation and status
- `sbatch`: Submits batch job scripts for later execution
- `srun`: Launches parallel jobs, either as standalone commands or within existing allocations
- `salloc`: Allocates resources for interactive use
- `scancel`: Cancels pending or running jobs

- `scontrol`: Administrative command for querying and modifying cluster configuration
- `sacct`: Retrieves accounting data for completed jobs

Each command supports numerous options and output format specifications, contributing to the system's flexibility but also its complexity. Understanding the full capabilities of these tools requires significant learning investment, motivating the development of more accessible interfaces.

2.1.4 Job Submission and Scheduling

Job submission in Slurm typically involves creating a batch script that specifies resource requirements and execution commands. The script uses special comment directives (prefixed with `#SBATCH`) to declare parameters such as partition, time limit, node count, and resource constraints.

Slurm's scheduler evaluates pending jobs and available resources to make allocation decisions. The default scheduler uses a priority-based algorithm with backfill optimization, where smaller jobs can be scheduled in gaps left by larger pending jobs. Administrators can configure various priority factors including job age, fair-share usage, job size, and quality-of-service levels.

The scheduling system supports advanced features including:

- Job arrays for submitting multiple similar jobs efficiently
- Job dependencies for defining execution order relationships
- Preemption policies for priority-based job interruption
- Resource reservations for guaranteed future allocations

These capabilities make Slurm suitable for complex workflows but increase the knowledge required for effective utilization.

2.2 Large Language Models

Large Language Models (LLMs) represent a class of artificial intelligence systems trained on vast corpora of text data to understand and generate human language [4]. These models have demonstrated remarkable capabilities in natural language understanding, reasoning, and generation, enabling a wide range of applications from conversational assistants to code synthesis. This section examines the fundamental concepts underlying LLMs and their relevance to the development of intelligent agents.

2.2.1 Transformer Architecture

Modern LLMs are built upon the Transformer architecture, introduced by Vaswani et al. in 2017. The Transformer revolutionized natural language processing by replacing recurrent neural network architectures with self-attention mechanisms, enabling efficient parallel processing of sequential data and capturing long-range dependencies in text.

The self-attention mechanism allows each token in a sequence to attend to all other tokens, computing relevance weights that determine how much information to gather from each position. This is expressed mathematically as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (2.1)$$

where Q , K , and V represent query, key, and value matrices derived from the input, and d_k is the dimension of the key vectors. The scaling factor $\sqrt{d_k}$ prevents the softmax function from entering regions of extremely small gradients.

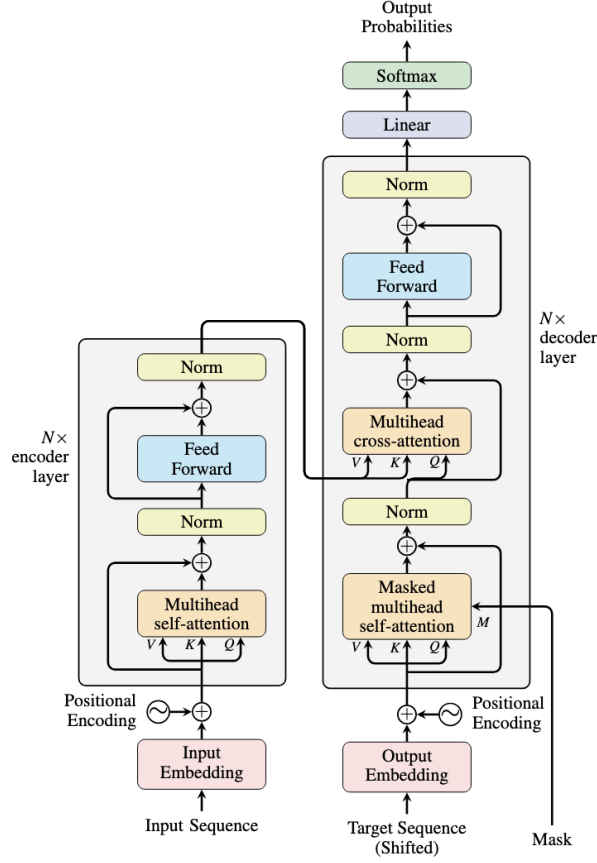


Figure 2.2: Transformer Architecture Overview

Transformers typically employ multi-head attention, where multiple attention computations are performed in parallel with different learned projections. This enables the model to capture different types of relationships and patterns simultaneously:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (2.2)$$

The architecture includes layer normalization, residual connections, and position-wise feed-forward networks that contribute to training stability and model expressiveness.

2.2.2 Autoregressive Language Modeling

LLMs are typically trained using autoregressive language modeling, where the model learns to predict the next token in a sequence given all preceding tokens. This training objective can be formalized as maximizing the log-likelihood:

$$\mathcal{L}(\theta) = \sum_{i=1}^n \log P(x_i | x_1, x_2, \dots, x_{i-1}; \theta) \quad (2.3)$$

where θ represents the model parameters and $x_1, \dots, x_n$ is the training sequence.

During inference, the model generates text by sampling or selecting tokens autoregressively, where each generated token becomes part of the context for generating subsequent tokens. Various decoding strategies exist, including:

- **Greedy decoding:** Always selecting the highest-probability token
- **Beam search:** Maintaining multiple candidate sequences simultaneously
- **Sampling:** Probabilistically selecting tokens based on the predicted distribution
- **Top-k/Top-p sampling:** Restricting sampling to the most likely candidates

2.2.3 Instruction Tuning and Alignment

While base language models are trained purely on next-token prediction, they require additional training to become useful assistants. Instruction tuning fine-tunes models on datasets of instructions paired with desired responses, enabling the model to follow user directives effectively.

Reinforcement Learning from Human Feedback (RLHF) further aligns models with human preferences [3]. This process involves:

1. Training a reward model on human preference data
2. Using the reward model to provide training signal
3. Optimizing the language model policy using reinforcement learning algorithms

The alignment process helps models produce outputs that are helpful, harmless, and honest—critical properties for deployment in real-world applications.

2.2.4 Emergent Capabilities

As LLMs scale in parameters and training data, they exhibit emergent capabilities not explicitly trained for. These include:

In-context learning: The ability to learn new tasks from examples provided in the prompt, without parameter updates. By presenting a few demonstrations of the desired input-output mapping, the model can generalize to new inputs.

Chain-of-thought reasoning: When prompted appropriately, LLMs can generate intermediate reasoning steps that improve performance on complex problems. This capability is particularly important for multi-step tasks requiring logical deduction.

Tool use: Models can learn to interact with external tools and APIs by generating appropriate function calls or structured outputs. This capability enables LLMs to access real-time information, perform calculations, and interact with external systems.

2.2.5 Function Calling and Structured Output

Modern LLMs support function calling, where the model can output structured requests to invoke external tools. Given a set of available functions with specified parameter schemas, the model determines when to call functions and how to construct appropriate arguments.

The function calling paradigm follows this pattern:

1. The system prompt defines available functions with their descriptions and parameter schemas
2. The model receives a user query and determines if function calls are needed
3. The model outputs structured function call requests with appropriate arguments
4. The application executes the function and returns results to the model
5. The model incorporates results and continues generation

This capability is fundamental to building LLM-based agents, as it provides the mechanism by which language models can take actions in the world rather than merely generating text responses.

2.3 Model Context Protocol

The Model Context Protocol (MCP) is an open protocol developed by Anthropic that standardizes the connection between Large Language Models and external data sources, tools, and services [1, 2]. MCP provides a unified interface for exposing functionality to AI agents, enabling modular and reusable integrations that can be shared across different applications. This section examines the protocol architecture, communication patterns, and implementation considerations.

2.3.1 Protocol Overview

MCP addresses a fundamental challenge in building AI agents: the need to connect language models to diverse external systems while maintaining security, modularity, and ease of development. Prior to MCP, each integration required custom implementation, leading to fragmented ecosystems and duplicated effort.

The protocol defines a client-server architecture where:

- **MCP Hosts** are applications that embed AI capabilities and initiate connections to servers
- **MCP Clients** are protocol implementations within hosts that maintain server connections
- **MCP Servers** are lightweight programs that expose specific capabilities through the protocol

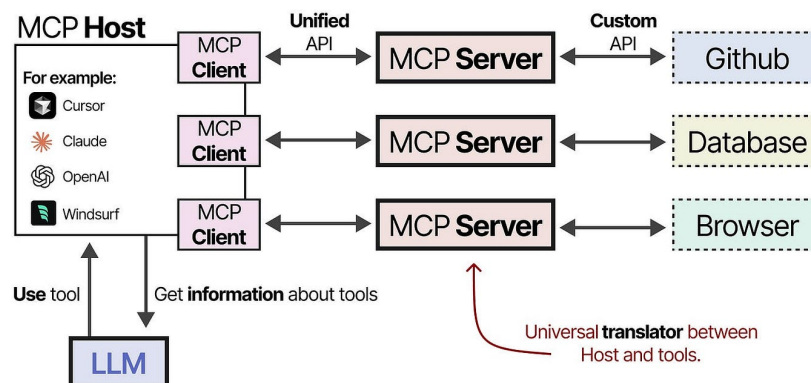


Figure 2.3: Model Context Protocol Architecture

MCP servers can expose three types of capabilities:

Resources provide data that can be loaded into the model's context. Resources are identified by URIs and can represent files, database records, API responses, or any other data the model might need to reference. Resources support subscriptions for change notifications.

Tools represent executable functions that the model can invoke to perform actions. Each tool has a defined name, description, and parameter schema that the model uses to determine when and how to call the tool. Tools are the primary mechanism for enabling agent actions.

Prompts are reusable templates that encapsulate specific interaction patterns. Servers can expose prompts that provide structured ways to interact with their capabilities, helping users leverage complex functionality effectively.

2.3.2 Transport Mechanisms

MCP supports multiple transport mechanisms for client-server communication, allowing flexibility in deployment scenarios:

Standard I/O (stdio) is the primary transport for local servers. The host spawns the server as a subprocess and communicates through standard input/output streams. This approach provides process isolation and is well-suited for command-line tools and local integrations.

HTTP with Server-Sent Events (SSE) enables remote server deployment. The client connects to the server over HTTP, with the server streaming responses using SSE. This transport supports network-based architectures and allows servers to be deployed as services.

The protocol uses JSON-RPC 2.0 as the message format, providing a standardized structure for requests, responses, and notifications. Messages include method names, parameters, and result data according to the JSON-RPC specification.

2.3.3 Tool Definition and Invocation

Tools are defined using JSON Schema to specify their parameter structure. A tool definition includes:

- A unique name identifying the tool
- A description explaining the tool's purpose and usage
- An input schema defining required and optional parameters

When a client connects, it queries available tools using the `tools/list` method. The server responds with the complete set of tool definitions, which the client provides to the language model as part of the function calling context.

Tool invocation follows this sequence:

1. The language model determines that a tool call is needed and generates the call with arguments
2. The client application receives the tool call request
3. The client invokes the tool on the MCP server using the `tools/call` method
4. The server executes the tool logic and returns results
5. The client passes results back to the language model for incorporation

2.3.4 Implementation Patterns

Implementing an MCP server involves creating handlers for protocol methods and defining the capabilities to expose. Common implementation patterns include:

Wrapper servers expose existing command-line tools or APIs through MCP. The server translates MCP tool calls into appropriate system commands or API requests, returning formatted results. This pattern enables rapid integration of existing functionality.

Aggregator servers combine multiple underlying services into a coherent capability set. The server manages connections to backend systems and presents a unified interface to the AI agent.

Stateful servers maintain context across tool invocations. While individual tool calls are stateless in the protocol, servers can maintain internal state for complex multi-step workflows.

Error handling in MCP follows standard patterns: servers return structured error responses with error codes and messages. Clients should handle both transport-level failures and application-level errors gracefully.

2.3.5 Security Considerations

MCP implementations must address several security concerns:

Input validation: All tool parameters must be validated before execution to prevent injection attacks or malformed inputs.

Capability restriction: Servers should expose only the minimum necessary functionality and implement appropriate access controls.

Output sanitization: Results returned to the language model should be sanitized to prevent information leakage or injection of malicious content.

Transport security: Remote deployments should use TLS encryption to protect data in transit.

The protocol itself does not mandate specific security mechanisms, leaving implementation decisions to developers based on their deployment context and requirements.

2.4 Agent Architectures and Frameworks

Building intelligent agents requires combining Large Language Models with structured reasoning patterns and tool integration frameworks. This section examines the ReAct agent architecture and the OpenAI Agents SDK, which form the foundation of the Slurm Agent implementation.

2.4.1 ReAct: Reasoning and Acting

The ReAct paradigm, introduced by Yao et al. [22], represents a significant advancement in LLM-based agent design. ReAct synergizes **reasoning** (generating verbal reasoning traces) with **acting** (taking actions that affect the environment), enabling agents to solve complex tasks that require both planning and external interaction.

The Think-Act-Observe Loop

ReAct agents operate through an iterative loop with three distinct phases:

1. **Think:** The agent generates a reasoning trace explaining its current understanding and intended next step. This internal monologue helps decompose complex problems and maintain task focus.
2. **Act:** Based on the reasoning, the agent selects and executes an action—typically invoking an external tool with specific parameters.
3. **Observe:** The agent receives the result of its action from the environment and incorporates this observation into its context for the next iteration.

This cycle repeats until the agent determines it has sufficient information to provide a final response, or encounters a termination condition.

ReAct Agent architecture

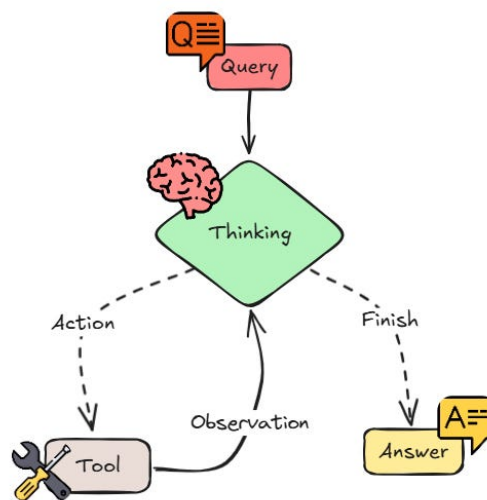


Figure 2.4: The ReAct Think-Act-Observe Loop

Advantages of ReAct

ReAct provides several advantages over pure reasoning or pure acting approaches:

Grounded Reasoning: By interleaving reasoning with actions, the agent can verify its hypotheses against real-world observations. This prevents the accumulation of errors that can occur in pure chain-of-thought reasoning.

Interpretable Decisions: The explicit reasoning traces make the agent’s decision-making process transparent, facilitating debugging and building user trust.

Dynamic Adaptation: Observations from actions can cause the agent to revise its plan, enabling adaptive behavior when initial assumptions prove incorrect.

Error Recovery: When an action fails or returns unexpected results, the agent can reason about the failure and attempt alternative approaches.

ReAct Prompt Structure

A ReAct agent is typically configured through prompting that establishes the think-act-observe pattern. The prompt structure includes:

- **System Instructions:** Define the agent’s role, available tools, and output format expectations
- **Tool Descriptions:** Specify each available tool’s purpose, parameters, and expected outputs
- **Few-shot Examples:** Demonstrate the desired reasoning and action patterns
- **Task Context:** Provide relevant background information for the current task

2.4.2 OpenAI Agents SDK

The OpenAI Agents SDK [11] is an open-source Python framework for building agentic AI applications. The SDK provides primitives for creating agents with tool-calling capabilities, managing agent handoffs, and implementing multi-agent systems.

Core Concepts

The SDK is built around several key abstractions:

Agent: An Agent encapsulates an LLM with specific instructions, available tools, and behavioral configurations. Agents can be created with custom system prompts that define their persona and capabilities.

Listing 2.1: Agent Definition Example

```

1 from agents import Agent, function_tool
2
3 @function_tool
4 def get_cluster_status() -> str:
5     """Get current Slurm cluster status."""
6     return execute_slurm_command("sinfo")
7
8 main_agent = Agent(
9     name="slurm_assistant",
10    instructions="You are a Slurm cluster management assistant...",
11    tools=[get_cluster_status],
12    model="gpt-4o"
13 )

```

Runner: The Runner executes agent workflows, managing the conversation loop, tool invocations, and response streaming. It handles the mechanics of sending messages to the LLM, processing tool calls, and collecting results.

Tools: Tools are functions that agents can invoke to interact with external systems. The SDK supports both synchronous and asynchronous tools, with automatic parameter validation based on function signatures and type hints.

Handoffs: Handoffs enable one agent to transfer control to another agent, passing along conversation context. This mechanism supports building multi-agent systems where specialized agents handle different aspects of a task.

Agent Execution Loop

When a Runner executes an agent, it follows this internal loop:

1. Collect all messages (system prompt, conversation history, user input)
2. Send to the LLM and receive a response
3. If the response contains tool calls:
 - (a) Execute each tool with the provided arguments
 - (b) Append tool results to the message history

- (c) Return to step 1 (continue the loop)
- 4. If the response is a final text message, return it to the caller
- 5. If the response is a handoff, transfer to the target agent

This loop naturally implements the ReAct pattern: the LLM's reasoning is embedded in its response generation, tool calls represent actions, and tool results provide observations.

Multi-Agent Handoffs

The SDK's handoff mechanism enables sophisticated multi-agent architectures. An agent can be configured as a handoff target, and other agents can transfer control using a special handoff tool:

Listing 2.2: Multi-Agent Handoff Configuration

```
1 analysis_agent = Agent(  
2     name="analysis_agent",  
3     instructions="Analyze_cluster_data_and_generate_insights...",  
4     tools=[analyze_cluster, generate_chart]  
5 )  
6  
7 main_agent = Agent(  
8     name="main_agent",  
9     instructions="Route_requests_to_appropriate_specialists...",  
10    handoffs=[analysis_agent]  
11 )
```

When the main agent determines that analysis is needed, it can invoke a handoff, transferring the conversation to the analysis agent. The target agent then handles the request and can either respond directly or hand back to the original agent.

Streaming Responses

The SDK supports streaming responses through the `Runner.run_streamed()` method, which yields events as they occur:

- **ResponseContentDelta:** Incremental text content from the agent
- **ToolCallStart:** Notification that a tool is being invoked
- **ToolCallComplete:** Tool execution result
- **AgentHandoff:** Transfer to another agent
- **ResponseComplete:** Final response ready

This streaming capability is essential for providing responsive user experiences, as users can see the agent’s thinking process in real-time rather than waiting for complete responses.

Context Management

The SDK provides mechanisms for managing conversation context:

RunContext: A context object passed to all tool functions, enabling tools to access shared state, configuration, and cross-cutting concerns.

Message History: Automatic tracking of the conversation, including user messages, agent responses, and tool interactions.

Model Context Protocol Integration: The SDK can connect to MCP servers, treating MCP tools as native agent tools through the mcp integration module.

2.4.3 Combining ReAct with OpenAI Agents SDK

The Slurm Agent system leverages both ReAct principles and the OpenAI Agents SDK infrastructure:

- **ReAct Reasoning:** The main agent’s instructions establish a ReAct-style reasoning pattern, encouraging the model to think through each request before taking action.
- **SDK Infrastructure:** The Agent and Runner classes provide the execution framework, handling tool invocations and response streaming.
- **MCP Tool Integration:** Slurm operations are exposed through an MCP server and connected to the agents via the SDK’s MCP integration.

- **Multi-Agent Coordination:** Specialized sub-agents handle analysis and action tasks, connected through the SDK's handoff mechanism.

This combination provides a robust foundation for building sophisticated AI agents that can reason about complex tasks, interact with external systems, and provide transparent, interpretable behavior.

Chapter 3

Related Tools and Works

3.1 HPC Management Tools

The management of High-Performance Computing clusters has traditionally relied on command-line interfaces and specialized monitoring solutions. This section reviews existing tools and approaches for HPC cluster management, analyzing their capabilities and limitations in the context of user accessibility.

3.1.1 Native Slurm Interfaces

Slurm itself provides the primary interface for cluster management through its command-line utilities. While powerful and comprehensive, these tools present several usability challenges:

sview is Slurm’s graphical user interface, providing visual representations of cluster state, job queues, and partition configurations. However, **sview** requires X11 forwarding for remote access, has limited customization options, and does not support advanced features such as natural language queries or intelligent recommendations. The interface provides visualization capabilities but lacks the flexibility of modern web-based solutions.

slurmrestd exposes Slurm functionality through a REST API, enabling programmatic access to cluster information and job management. While this opens possibilities for building custom interfaces, developers must still handle the complexity of Slurm’s

data models and operations. The REST interface provides machine-readable access but does not address user-facing accessibility concerns.

3.1.2 Third-Party Monitoring Solutions

Several third-party tools have emerged to provide enhanced cluster monitoring and management capabilities:

Open XDMoD (Open XD Metrics on Demand) is an open-source tool for comprehensive HPC resource management and reporting. It provides detailed analytics on resource utilization, job statistics, and user behavior. XDMoD excels at historical analysis and reporting but focuses on administrative oversight rather than user-facing job management.

Grafana with Prometheus represents a common monitoring stack deployed alongside Slurm clusters. Prometheus collects metrics from cluster nodes, while Grafana provides customizable dashboards for visualization. This combination offers flexible monitoring but requires significant configuration effort and does not provide interactive job management capabilities.

SLURM Job Viz and similar tools provide web-based visualization of cluster state. These tools typically offer graphical views of node allocation and job queues. While improving visibility, they remain read-only monitoring solutions without job submission or management functionality.

Table 3.1: Comparison of HPC Management Tools

Tool	NL Interface	Job Mgmt	Visualization	Web-based
Slurm CLI	No	Yes	No	No
sview	No	Yes	Yes	No
Open XDMoD	No	Limited	Yes	Yes
Grafana	No	No	Yes	Yes
Slurm Agent	Yes	Yes	Yes	Yes

3.1.3 Portal and Gateway Systems

Science gateways and web portals have been developed to provide more accessible interfaces to HPC resources:

Open OnDemand is a popular open-source platform providing web-based access to HPC clusters. It offers file management, job submission through web forms, and interactive application launching. While significantly improving accessibility over command-line interfaces, Open OnDemand still requires users to understand Slurm concepts and manually specify job parameters.

JupyterHub deployments on HPC systems enable interactive notebook-based computing. Users can launch Jupyter sessions that run on compute nodes, accessing cluster resources through a familiar interface. However, job management and cluster monitoring remain separate from the notebook environment.

These portal solutions improve accessibility but do not address the fundamental challenge of translating user intent into appropriate cluster operations. Users must still possess knowledge of resource requirements, partition configurations, and job specifications.

3.1.4 Limitations of Existing Approaches

Analysis of existing HPC management tools reveals common limitations that motivate the development of intelligent agent-based interfaces:

Knowledge requirements: All existing tools assume users understand Slurm concepts, command syntax, or form field meanings. There is no mechanism for translating high-level goals into appropriate system operations.

Static interfaces: Current tools provide fixed interfaces that cannot adapt to individual user needs or provide contextual guidance based on the current situation.

Limited reasoning: Existing solutions cannot reason about cluster state to provide recommendations, diagnose issues, or suggest optimizations.

No natural language: Users must communicate through structured interfaces (commands, forms, queries) rather than expressing needs in natural language.

These limitations create opportunities for LLM-based agents that can understand user

intent, reason about cluster operations, and provide intelligent assistance—the approach pursued in this research.

3.2 LLM-Based Agent Frameworks

The development of LLM-based agents has accelerated rapidly, with numerous frameworks emerging to facilitate the creation of AI systems that can reason, plan, and take actions. This section surveys relevant agent frameworks and patterns that inform the design of the Slurm Agent system.

3.2.1 Agent Paradigms

LLM-based agents extend language models beyond text generation to systems capable of interacting with external environments. Several paradigms have emerged:

ReAct (Reason + Act) combines reasoning and acting in an interleaved manner. The agent alternates between generating reasoning traces (thoughts) and taking actions, using observations from actions to inform subsequent reasoning. This pattern enables transparent decision-making and effective problem decomposition.

Chain-of-Thought (CoT) prompting encourages models to generate intermediate reasoning steps before producing final answers. When combined with tool use, CoT enables agents to plan multi-step solutions and verify their progress.

Reflexion introduces self-reflection mechanisms where agents evaluate their own outputs and learn from mistakes within a single episode. Failed attempts generate feedback that informs subsequent tries, improving performance without parameter updates.

Tree-of-Thoughts (ToT) extends linear reasoning to explore multiple reasoning paths simultaneously. The agent evaluates alternative approaches and can backtrack when one path proves unproductive.

3.2.2 Framework Implementations

Several open-source frameworks provide infrastructure for building LLM-based agents:

ATTRIBUTE	LANGGRAPH	LANGCHAIN	MICROSOFT AUTOGEN	CREWAI	OPENAI AGENTS SDK
USAGE	Complex stateful agents, iterative workflows	General-purpose LLM app building, RAG, agents	Multi-agent conversations, automation, code exec.	Role-based multi-agent collaboration, task teams	Production-ready agents, handoffs, guardrails
COMPLEXITY	High	Medium to High	Medium to High	Low to Medium	Medium
LEARNING CURVE	High	Medium	Medium	Low	Low to Medium
ARCHITECTURE	Graph-based (nodes, edges, stateful cycles)	Modular chains, agents, tools	Conversational, multi-agent, flexible topologies	Role-based crew, sequential/parallel task flow	Agent primitives, handoffs, guardrails
SPEED	Can be highly optimized for iterative processes	Depends on chain complexity and LLM calls	Concurrent conversations, can be efficient	Efficient for coordinated tasks	Designed for fast execution, production-focused
SCALABILITY	Excellent (explicit state, robust for production)	Good (with appropriate infrastructure)	Good (designed for distributed agent networks)	Good (for multi-agent coordination)	Good (built for production workloads)
MAINTAINABILITY	High (explicit graph structure aids debugging)	Moderate (can become complex with large apps)	Moderate (clear agent roles aid maintenance)	High (clear roles and task definitions)	High (clear primitives, strong tracing)

Figure 3.1: Comparison of LLM Agent Frameworks

3.2.3 Multi-Agent Systems

Complex tasks often benefit from multiple specialized agents working together. Multi-agent architectures divide responsibilities among agents with distinct capabilities:

Hierarchical architectures employ a coordinator agent that delegates subtasks to specialized worker agents. The coordinator maintains overall task state and synthesizes results from workers.

Collaborative architectures enable peer agents to communicate and coordinate without central control. Agents may negotiate, share information, or critique each other's outputs.

Handoff patterns allow one agent to transfer control to another when the task requires different capabilities. The receiving agent continues execution with relevant context from the handoff.

The OpenAI Agents SDK specifically supports handoff patterns, where agents can transfer conversations to other agents based on task requirements. This mechanism enables modular agent design where each agent focuses on specific capabilities.

3.2.4 Tool Integration Patterns

Effective agent systems require robust tool integration. Common patterns include:

Direct function calling: Tools are defined as functions that the LLM can invoke directly. This approach is simple but couples tool definitions to the agent implementation.

Plugin architectures: Tools are packaged as plugins that can be dynamically loaded and configured. This enables extensibility without modifying core agent code.

Protocol-based integration: Tools are exposed through standardized protocols (such as MCP), enabling interoperability across different agent frameworks and tool providers.

The protocol-based approach offers significant advantages for maintainability and reusability. Tools developed for one agent system can be used with others, and tool implementations can evolve independently of agent logic.

3.2.5 Relevance to Slurm Agent

The Slurm Agent design draws on lessons from existing frameworks:

- The ReAct pattern informs the agent’s approach to reasoning about Slurm operations
- Multi-agent architecture with handoffs enables specialized handling of different task types
- The OpenAI Agents SDK provides the implementation foundation
- MCP-based tool integration ensures modularity and security

By combining proven patterns with domain-specific design, the Slurm Agent addresses the unique requirements of HPC cluster management while leveraging established best practices from the agent development community.

3.3 MCP Implementations and Applications

Since its introduction, the Model Context Protocol has seen growing adoption across the AI development community. This section reviews notable MCP implementations and applications, examining how the protocol is being used to connect AI agents with external systems.

3.3.1 Reference Implementations

Anthropic has released reference implementations and SDKs to facilitate MCP adoption:

Python MCP SDK provides a comprehensive library for building MCP servers and clients in Python. The SDK handles protocol details including transport management, message serialization, and connection lifecycle. It offers decorators for defining tools, resources, and prompts with minimal boilerplate.

TypeScript MCP SDK provides equivalent functionality for JavaScript/TypeScript environments. This enables building MCP servers as Node.js applications and integrating MCP clients into web and desktop applications.

These SDKs significantly reduce the effort required to implement MCP integrations, enabling developers to focus on capability logic rather than protocol mechanics.

3.3.2 Server Ecosystem

A growing ecosystem of MCP servers exposes various system capabilities:

Filesystem servers provide access to local or remote file systems, enabling agents to read, write, and navigate files. These servers implement appropriate sandboxing to control what paths agents can access.

Database servers expose database query and manipulation capabilities. Servers have been developed for PostgreSQL, SQLite, and other database systems, allowing agents to retrieve data and perform operations.

API integration servers connect agents to external services. Examples include servers for GitHub (repository management), Slack (messaging), and various cloud platforms. These servers translate MCP tool calls into appropriate API requests.

Development tool servers integrate with software development workflows. Servers exist for code search, test execution, and build systems, enabling AI-assisted software development.

3.3.3 Host Applications

Several applications have integrated MCP client capabilities:

Claude Desktop application natively supports MCP, enabling users to connect Claude to local and remote MCP servers. This demonstrates the end-user experience of MCP-enabled AI assistants.

Development environments are integrating MCP for AI-assisted coding. IDE extensions and editor plugins can leverage MCP servers to provide contextual assistance based on project files and developer tools.

Custom applications built using MCP client libraries can connect to any compatible server. This flexibility enables domain-specific AI applications that leverage shared tool ecosystems.

3.3.4 Design Patterns in Practice

Analysis of existing MCP implementations reveals common design patterns:

Wrapper pattern: Many servers wrap existing command-line tools or APIs. The server parses tool inputs, executes underlying commands, and formats outputs for consumption by language models. This pattern enables rapid integration of existing functionality.

Composition pattern: Servers may compose multiple underlying capabilities into coherent tool sets. For example, a development server might combine file access, Git operations, and build commands into a unified interface.

Filtering pattern: Servers implement filtering to control what information is exposed to agents. Database servers may limit queryable tables; filesystem servers may restrict accessible paths. This pattern addresses security and scope concerns.

Formatting pattern: Outputs are formatted to be LLM-friendly, often using markdown or structured formats that models can effectively interpret. Clear, consistent output formatting improves agent reliability.

3.3.5 Gaps and Opportunities

Despite the growing ecosystem, several gaps remain:

HPC domain: Prior to this research, no MCP servers specifically targeted HPC workload management. The complexity and specialized nature of systems like Slurm require domain-specific tool design.

Visualization: While servers can return data, few integrate visualization generation capabilities that enable rich information display within agent interfaces.

Multi-agent coordination: Most MCP applications use single-agent architectures. Patterns for using MCP tools across multi-agent systems remain underexplored.

The Slurm Agent addresses these gaps by implementing an MCP server tailored for HPC management, integrating visualization capabilities, and demonstrating multi-agent architectures with MCP-based tool access. This contribution extends the MCP ecosystem into the high-performance computing domain.

Chapter 4

Proposed Approaches

4.1 Application Introduction

The Slurm Agent is an intelligent conversational interface designed to democratize access to High-Performance Computing cluster management. By combining Large Language Models with the Model Context Protocol, the system enables users to interact with Slurm workload managers through natural language queries, eliminating the need to memorize complex command syntax and reducing the cognitive overhead associated with HPC operations.

4.1.1 Application Concept

The fundamental concept behind the Slurm Agent is to create an intermediary layer between users and the Slurm cluster that understands natural language, reasons about appropriate actions, and presents results in accessible formats. Users express their needs conversationally—such as “What jobs are running on the GPU partition?” or “Show me a chart of my job history”—and the agent translates these queries into appropriate Slurm commands, executes them, and presents formatted results.

The system implements a **ReAct (Reasoning and Acting)** architecture pattern, where the main agent follows a Think → Act → Observe loop to process user requests. This enables self-correction on failures and iterative refinement of responses based on tool execution results.

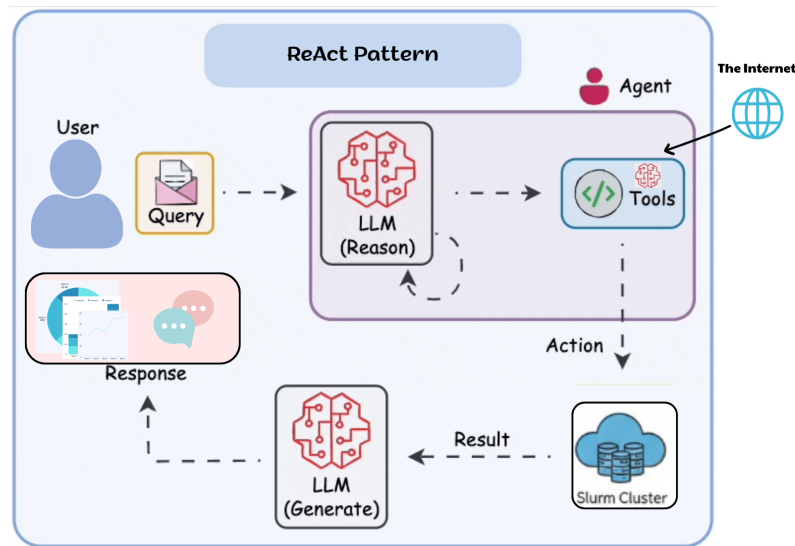


Figure 4.1: Slurm Agent Application Concept

The application provides several key capabilities:

Natural language query processing: Users can ask questions about cluster state, job status, and resource availability using conversational language. The main agent interprets intent and delegates to specialized sub-agents for data gathering.

Multi-agent orchestration: A main ReAct agent orchestrates specialized sub-agents—an Analysis Sub-Agent for read-only queries (cluster status, job information, web search) and an Action Sub-Agent for job management operations (submit, cancel, hold, release).

Command execution with safety controls: The system distinguishes between safe read-only operations and dangerous write operations. Dangerous actions (job cancellation, configuration changes) are queued and require explicit user confirmation before execution.

Data visualization: Cluster metrics and job statistics can be visualized as charts and graphs through a dedicated chart generation tool, providing intuitive understanding of complex data.

Documentation assistance: When users encounter errors or need guidance, the agent can search relevant documentation via web search integration and provide contextual help.

Real-time streaming: Responses stream to users in real-time, providing immediate feedback during processing and maintaining a responsive user experience.

4.1.2 Target Users

The application serves multiple user categories with different needs:

Researchers and scientists who use HPC resources for computational work but may lack deep system administration knowledge. These users benefit from the ability to check job status, understand resource availability, and troubleshoot issues without mastering Slurm’s command-line interface.

New HPC users face a steep learning curve when first accessing cluster resources. The Slurm Agent reduces this barrier by enabling productive interaction from the start, with the agent handling translation to system operations.

System administrators can use the agent for rapid cluster status checks and routine operations, leveraging natural language for efficiency. The agent supplements rather than replaces direct command-line access.

Support staff helping users with HPC-related questions can use the agent to quickly gather diagnostic information and provide guided assistance.

4.1.3 Interaction Model

User interaction with the Slurm Agent follows a conversational paradigm. Users type messages in a chat interface, and the agent responds with text, formatted data, or visualizations. The conversation maintains context across multiple turns through persistent session management, allowing follow-up questions and clarifications.

A typical interaction flow includes:

1. User submits a natural language query through the web interface
2. Main ReAct agent reasons about the query (*Think* phase)
3. Agent determines necessary actions and invokes appropriate sub-agents (*Act* phase)
4. Sub-agents call tools to execute Slurm commands
5. Tool results are observed and processed (*Observe* phase)
6. For dangerous operations, action is queued pending user confirmation
7. Agent generates a natural language response with formatted results

8. Visualizations, if requested, are rendered inline as charts
9. User can continue with follow-up queries using preserved context

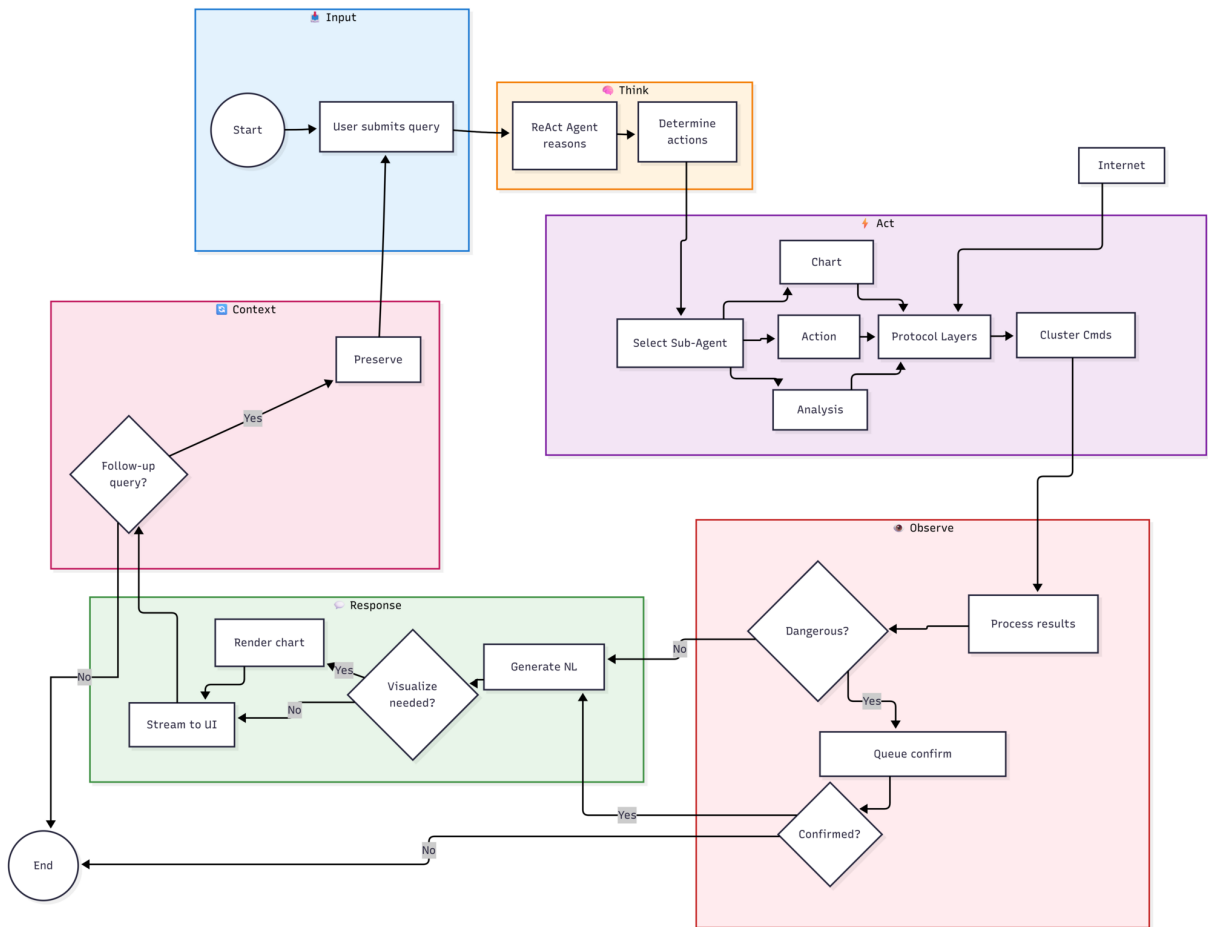


Figure 4.2: Slurm Agent Flow Chart

This model enables intuitive interaction while maintaining the full power of Slurm’s capabilities through the underlying tool integration, with safety controls preventing accidental cluster modifications.

4.2 Requirements Overview

This section presents the functional and non-functional requirements that guide the design and implementation of the Slurm Agent system. Requirements are derived from the project goals and the needs of target users identified in the previous section.

4.2.1 Functional Requirements

Functional requirements define the specific capabilities the system must provide:

FR1: Natural Language Understanding. The system shall interpret user queries expressed in natural language and determine appropriate actions. This includes handling variations in phrasing, implicit context, and ambiguous requests through clarification dialogue.

FR2: Slurm Query Execution. The system shall execute Slurm queries to retrieve cluster information, including:

- Partition information (names, states, resource limits)
- Node status (availability, resources, current allocations)
- Job queue status (pending and running jobs)
- Job details (parameters, resources, timing information)
- Accounting data (historical job information)

FR3: Job Management Operations. The system shall support job management actions including:

- Job submission via batch scripts
- Job cancellation for running or pending jobs
- Job control operations (hold, release, modify priority)
- Retrieving detailed job information and output files

FR4: Data Visualization. The system shall generate visualizations for cluster data, including:

- Node resource utilization charts
- Job distribution by partition
- Historical usage trends
- Queue statistics

FR5: Web Search Integration. The system shall search web resources to assist with troubleshooting and documentation lookup, enabling access to current information beyond the system's training data.

FR6: Real-Time Response Streaming. The system shall stream responses in real-time, providing incremental updates as the agent processes queries and generates results.

FR7: Markdown Response Formatting. The system shall format responses using markdown, enabling rich text display including tables, code blocks, and inline formatting.

FR8: Multi-Turn Conversation. The system shall maintain conversation context across multiple turns, enabling follow-up questions and contextual references.

FR9: Confirmation for Dangerous Operations. The system shall require explicit user confirmation before executing operations that modify cluster state, such as job cancellation or configuration changes. Pending actions shall be stored persistently until confirmed or cancelled.

4.2.2 Non-Functional Requirements

Non-functional requirements define quality attributes and constraints:

NFR1: Response Latency. The system should begin streaming responses within 2 seconds of query submission. Complete responses for simple queries should typically complete within 10 seconds.

NFR2: Accuracy. Slurm command outputs should be accurately retrieved and presented. The agent should not fabricate cluster information.

NFR3: Safety. The system shall not execute commands that could harm cluster operations. Dangerous operations shall be intercepted by a confirmation mechanism requiring explicit user approval before execution.

NFR4: Usability. The web interface shall be intuitive and usable without specialized training. Users should be able to begin productive interaction immediately.

NFR5: Extensibility. The MCP server architecture shall support addition of new tools without modifying the agent core. Visualization capabilities shall be extensible through modular chart definitions.

NFR6: Maintainability. Code shall be organized in modular components with clear

separation of concerns. The multi-agent architecture shall isolate different functional domains.

Table 4.1: Requirements Traceability to Project Goals

Requirement	G1 <i>NL Interface</i>	G2 <i>Multi- Agent</i>	G3 <i>MCP Tools</i>	G4 <i>Visual- ization</i>	G5 <i>Stream- ing</i>	G6 <i>Web UI</i>
FR1: NL Understanding	✓					
FR2: Query Execution	✓		✓			
FR3: Job Management	✓	✓	✓			
FR4: Visualization			✓	✓		
FR5: Web Search		✓	✓			
FR6: Streaming					✓	✓
FR7: Formatting						✓
FR8: Multi-Turn						✓
FR9: Confirmation		✓				✓

4.2.3 Constraints

The following constraints bound the system implementation:

Technology constraints: The system uses the OpenAI API for LLM capabilities, requiring API access and incurring usage costs. The MCP server is implemented in Python using the official SDK.

Environment constraints: Testing is conducted on a single-node Slurm cluster configuration. The system assumes standard Slurm command availability.

Scope constraints: The current implementation focuses on read operations and visualization. Administrative write operations are limited to prevent accidental cluster impact.

4.3 System Architecture

This section presents the overall architecture of the Slurm Agent system, describing the major components, their interactions, and the rationale behind architectural decisions.

The architecture follows a layered design that separates user interface, agent logic, and system integration concerns.

4.3.1 Architecture Overview

The Slurm Agent system consists of four main layers:

1. **Presentation Layer:** Web-based frontend for user interaction
2. **Agent Layer:** Multi-agent backend handling reasoning and orchestration
3. **Protocol Layer:** MCP server providing standardized tool interfaces
4. **Infrastructure Layer:** Slurm cluster and external services

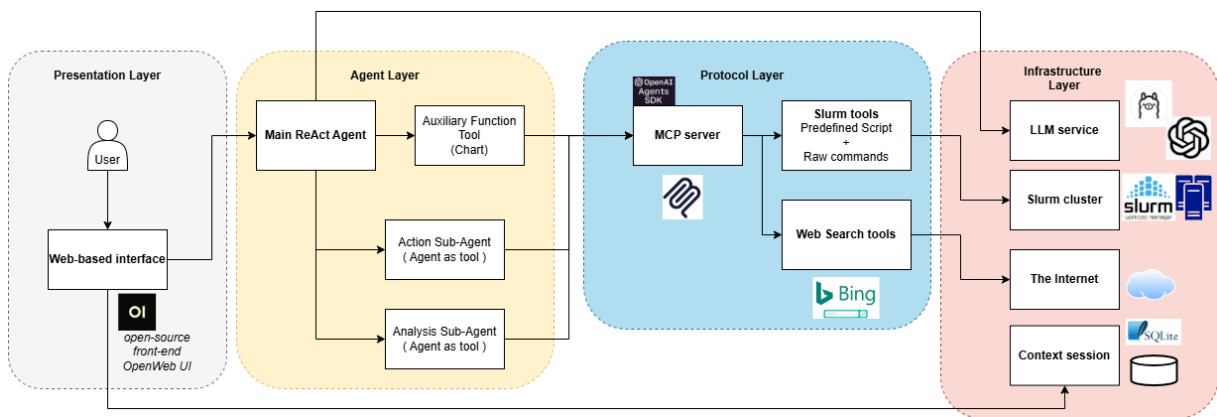


Figure 4.3: Slurm Agent System Architecture

4.3.2 Component Descriptions

Presentation Layer

The Presentation Layer provides the user-facing interface for interacting with the Slurm Agent. This layer is responsible for:

- Rendering a conversational chat interface with message history
- Capturing user queries in natural language
- Displaying streaming responses in real-time

- Rendering formatted content including markdown, code blocks, and inline visualizations
- Managing confirmation workflows for dangerous operations
- Preserving session state across interactions

The interface follows a conversational paradigm where users type queries and receive responses in a familiar messaging format. This design abstracts the complexity of Slurm commands behind natural language interaction.

Agent Layer

The Agent Layer implements the core reasoning and orchestration logic using a multi-agent architecture. The system employs a hierarchical structure with specialized agents:

- **Main ReAct Agent:** The primary orchestrator that implements the Think $\rightarrow$ Act $\rightarrow$ Observe reasoning loop. This agent maintains conversation context, interprets user intent, and delegates to specialized sub-agents based on the nature of the request.
- **Analysis Sub-Agent:** Handles read-only information gathering operations including cluster status queries, job queue inspection, resource monitoring, and documentation search. This agent can chain multiple tool calls to gather comprehensive information.
- **Action Sub-Agent:** Manages operations that modify cluster state, such as job submission, cancellation, and configuration changes. This agent enforces safety controls and confirmation requirements for dangerous operations.

The Main Agent also has direct access to auxiliary tools including chart generation for data visualization, which calls the Protocol Layer to render cluster metrics and statistics as visual charts.

Key architectural features of the Agent Layer include:

- **Session Persistence:** Conversation history is maintained across interactions, enabling contextual follow-up queries

- **Confirmation Workflow:** Dangerous operations are queued pending explicit user confirmation before execution
- **Token Optimization:** Visualization artifacts are filtered from conversation history to reduce token consumption
- **Streaming Output:** Responses are streamed incrementally to provide immediate user feedback

Protocol Layer

The Protocol Layer implements the Model Context Protocol (MCP) to provide a standardized interface between the Agent Layer and the underlying Slurm infrastructure. This layer:

- **Exposes Slurm Operations as Tools:** Each Slurm command is wrapped as a discrete tool with defined parameters and return types
- **Validates Parameters:** Input validation prevents malformed or dangerous command construction
- **Executes Commands Safely:** Commands are executed with appropriate isolation and error handling
- **Formats Output:** Raw command output is processed into structured formats suitable for agent consumption

The MCP abstraction provides several benefits:

- Decouples agent logic from Slurm-specific implementation details
- Enables tool discovery and introspection by the agent
- Supports extension with new tools without modifying the agent core
- Provides a security boundary for command execution

Infrastructure Layer

The Infrastructure Layer comprises the external systems that the agent interacts with:

- **Large Language Model:** Provides the reasoning capabilities for natural language understanding, intent interpretation, and response generation (either local or cloud API)
- **Slurm Cluster:** The target HPC infrastructure being managed, accessed through standard Slurm commands
- **Web Search Service:** Enables retrieval of documentation and troubleshooting information from external sources
- **Context Database:** SQLite database for storing session state and pending confirmation actions

4.3.3 Communication Patterns

The architecture employs several communication patterns:

Request-Response: Standard requests for initiating conversations and submitting queries.

Streaming: Unidirectional streaming from backend to frontend for real-time response delivery. The backend sends a continuous stream of events that the frontend processes incrementally.

Tool Invocation: Structured communication between the Agent Layer and Protocol Layer for tool discovery and execution.

4.3.4 Data Flow

A typical query flows through the system layers as follows:

1. User submits natural language query through the Presentation Layer
2. Agent Layer receives query and begins Think-Act-Observe reasoning
3. Agent delegates to appropriate sub-agent based on query type

4. Sub-agent invokes tools via Protocol Layer to gather data or perform actions
5. Results propagate back through the layers
6. Agent synthesizes response and streams to user

For dangerous operations, an additional confirmation step interrupts the flow: the pending action is stored and presented to the user, and execution proceeds only after explicit confirmation.

4.3.5 Deployment Architecture

The system components can be deployed in various configurations:

Local Development: All components run on a single machine with local Slurm installation or connection to a remote cluster.

Distributed Deployment: Frontend, backend, and MCP server run on separate hosts, with the MCP server deployed on a Slurm management node.

The modular architecture supports flexible deployment topologies based on infrastructure requirements and security considerations.

Chapter 5

Implementation

5.1 Agent Backend Implementation

This section describes the implementation of the agent backend, which orchestrates the multi-agent system and manages communication between the frontend, LLM API, and MCP server. The design decisions presented here reflect careful consideration of safety, usability, and architectural clarity in the context of HPC cluster management.

5.1.1 Technology Stack

The backend is implemented using Python 3.10+ as the primary language, selected for its extensive ecosystem of AI and asynchronous programming libraries. The core technology stack includes:

- **OpenAI Agents SDK** [11]: Foundational framework for agent implementation, offering built-in support for tool calling, conversation management, and multi-agent orchestration
- **FastAPI** (v0.116+) [12]: High-performance asynchronous web framework enabling efficient handling of concurrent requests
- **Uvicorn**: ASGI server for production deployment
- **Pydantic**: Structured output validation and command parameter modeling

- **SQLite**: Persistent storage for session management, conversation history, and pending actions
- **MCP Client Library**: Establishes the connection to the MCP server via SSE transport [7]
- **SSE-Starlette**: Provides streaming infrastructure for real-time response delivery to the frontend in OpenWebUI-compatible format

5.1.2 Multi-Agent Architecture: Sub-Agents as Tools

A critical architectural decision in this implementation is the use of specialized sub-agents implemented as callable tools rather than as simple function-based tools [20]. This design choice addresses several challenges inherent in HPC cluster management operations.

Rationale for Sub-Agent Design

Traditional tool-based architectures assume that each tool performs a discrete, well-defined operation with predictable input-output patterns [16]. However, Slurm cluster operations frequently require complex reasoning chains that simple tools cannot adequately support. For example, analyzing job failures requires not only retrieving job data via `sacct`, but also correlating that data with node status from `sinfo`, examining recent configuration changes, and potentially searching documentation for error codes. This multi-step reasoning process exceeds the capabilities of a single function tool.

By implementing the Analysis Sub-Agent and Action Sub-Agent as dedicated agents rather than simple functions, the system gains several advantages. First, each sub-agent maintains its own instruction set optimized for its specific domain, enabling focused reasoning without the cognitive overhead of a monolithic prompt. Second, sub-agents can autonomously chain multiple MCP tool calls within a single invocation, performing iterative data gathering without returning to the main agent after each step. Third, tool isolation prevents the main agent from being overwhelmed by the full complexity of all available Slurm commands, instead presenting a simplified interface of “analyze the cluster” and “manage jobs.”

Tool Access Control

A key architectural feature is strict tool isolation between agents. Table 5.1 shows the complete tool access matrix.

Table 5.1: Agent Tool Access Matrix

Tool	Main	Analysis	Action	Confirmation
<i>Query Tools</i>				
squeue		✓	✓	No
sacct		✓	✓	No
sinfo		✓	✓	No
scontrol_show		✓	✓	No
run_analysis		✓		No
web_search		✓		No
sacctmgr_show			✓	No
sreport			✓	No
<i>Action Tools</i>				
sbatch			✓	Yes
scancel			✓	Yes
scontrol_hold			✓	Yes
scontrol_release			✓	Yes
scontrol_update			✓	Yes
scontrol_requeue			✓	Yes
srun			✓	No
salloc			✓	No
<i>Admin Tools</i>				
scontrol_create/delete			✓	Yes
scontrol_reconfigure			✓	Yes
sacctmgr_add/modify/delete			✓	Yes
<i>Main Agent Direct Tools</i>				
analyze_cluster (sub-agent)	✓			No
manage_jobs (sub-agent)	✓			No
generate_chart	✓			No
confirm_action	✓			No
cancel_action	✓			No

Analysis Sub-Agent Specification

The Analysis Sub-Agent is designated for read-only data gathering operations. When the main agent determines that cluster investigation is needed, it invokes the Analysis Sub-Agent with a natural language request such as “investigate why jobs in partition gpu are pending.” The sub-agent autonomously determines which tools to call, chains multiple queries if needed, and returns a structured summary for synthesis into a user-facing response.

Action Sub-Agent Specification

The Action Sub-Agent handles operations that modify cluster state. The segregation of action-oriented operations into a dedicated sub-agent serves both safety and clarity purposes. The Action Sub-Agent’s instructions emphasize verification requirements and consequence awareness, ensuring that potentially destructive operations receive appropriate scrutiny before execution. Dangerous operations are intercepted by the confirmation mechanism before reaching actual execution.

Agent Prompt Engineering

Each agent in the multi-agent system has carefully crafted instruction prompts that define its behavior, capabilities, and output format. These prompts are critical for ensuring reliable operation within the HPC cluster management context.

Main Agent Prompt Design The main agent serves as the orchestrator and user-facing interface. Its prompt is structured into several key sections:

- **Critical Rules:** Establishes behavioral constraints, most importantly the rule to “ALWAYS call tools first” rather than fabricating information. This prevents hallucination of cluster status.
- **Tool Descriptions:** Documents available tools including `analyze_cluster` for data gathering, `manage_jobs` for actions, and `generate_chart` for visualization.

- **Output Format Specification:** Defines structured table formats for presenting cluster status, job landscapes, and recommendations. This ensures consistent, scannable output.
- **Workflow Guidelines:** Provides step-by-step guidance for common scenarios such as web searches for debugging and cluster data analysis.

Analysis Sub-Agent Prompt Design The Analysis Sub-Agent uses a minimal, focused prompt optimized for tool execution rather than conversation:

- **Single-Tool Invocation:** The prompt explicitly states “Call ONE tool only, then stop,” preventing unnecessary reasoning loops.
- **Semantic Tool Selection:** The prompt guides tool selection based on user intent rather than keyword patterns. Requests seeking external knowledge (troubleshooting, error explanations, documentation) route to the web search tool, while data-gathering requests route to analysis scripts.
- **Raw Output Policy:** Instructions to “Return raw tool output - do not summarize” ensures the main agent receives complete data for synthesis.

Action Sub-Agent Prompt Design The Action Sub-Agent prompt emphasizes operational precision:

- **Tool Categorization:** Groups tools by function (job control, interactive, admin, accounting) for clarity.
- **Confirmation Awareness:** Explicit mention that “Dangerous actions are queued for user confirmation” ensures the sub-agent understands the two-factor flow.
- **Result Reporting:** Instruction to “Return execution results” maintains transparency for user feedback.

Model Selection Strategy The implementation uses different models for different agent roles:

- **Reasoning Model:** Used by the main agent with higher temperature (0.7) for creative problem-solving and natural conversation.
- **Tool Model:** Used by sub-agents with low temperature (0.1) for deterministic, precise tool execution.

This separation ensures that user-facing responses benefit from natural language fluency while tool invocations remain reliable and predictable.

The complete query processing workflow, including the Think-Act-Observe loop, sub-agent invocation, and confirmation handling, is illustrated in Figure 5.1.

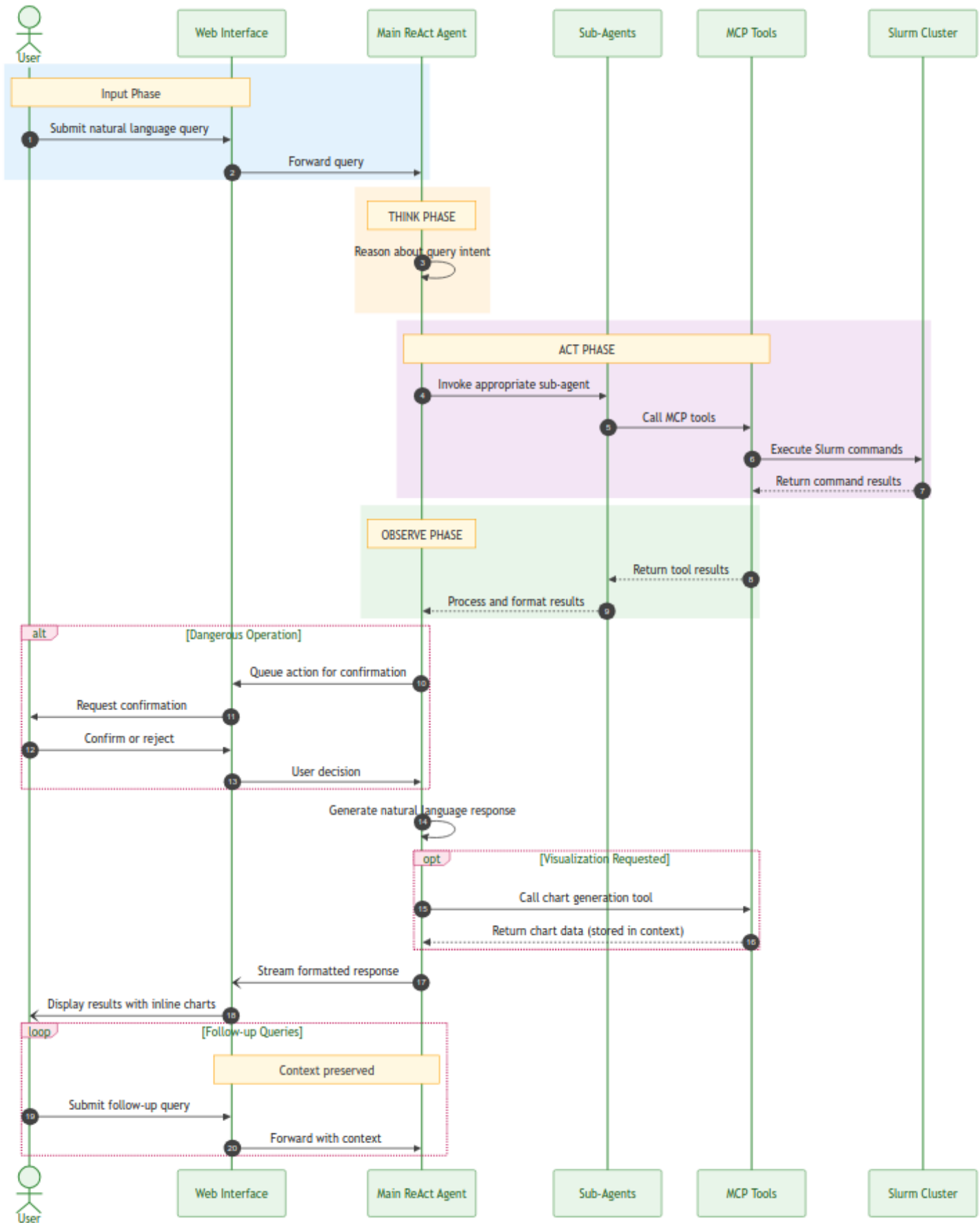


Figure 5.1: Query Processing Sequence

5.1.3 Confirmation Context: Two-Factor Authentication for Dangerous Operations

The confirmation context mechanism prevents accidental execution of destructive operations. When a dangerous tool (e.g., `scancel`, `scontrol_update`) is invoked, the system stores the pending action in SQLite rather than executing immediately, then returns a confirmation request to the user. This mirrors two-factor authentication: the first factor is the user's request, and the second is explicit confirmation.

Pending actions persist across session boundaries with automatic expiration after one hour. The confirmation tools (`confirm_action` and `cancel_action`) are available to the main agent, enabling natural conversational flow while enforcing the two-step verification process. The confirmation flow is illustrated in Figure 5.2.

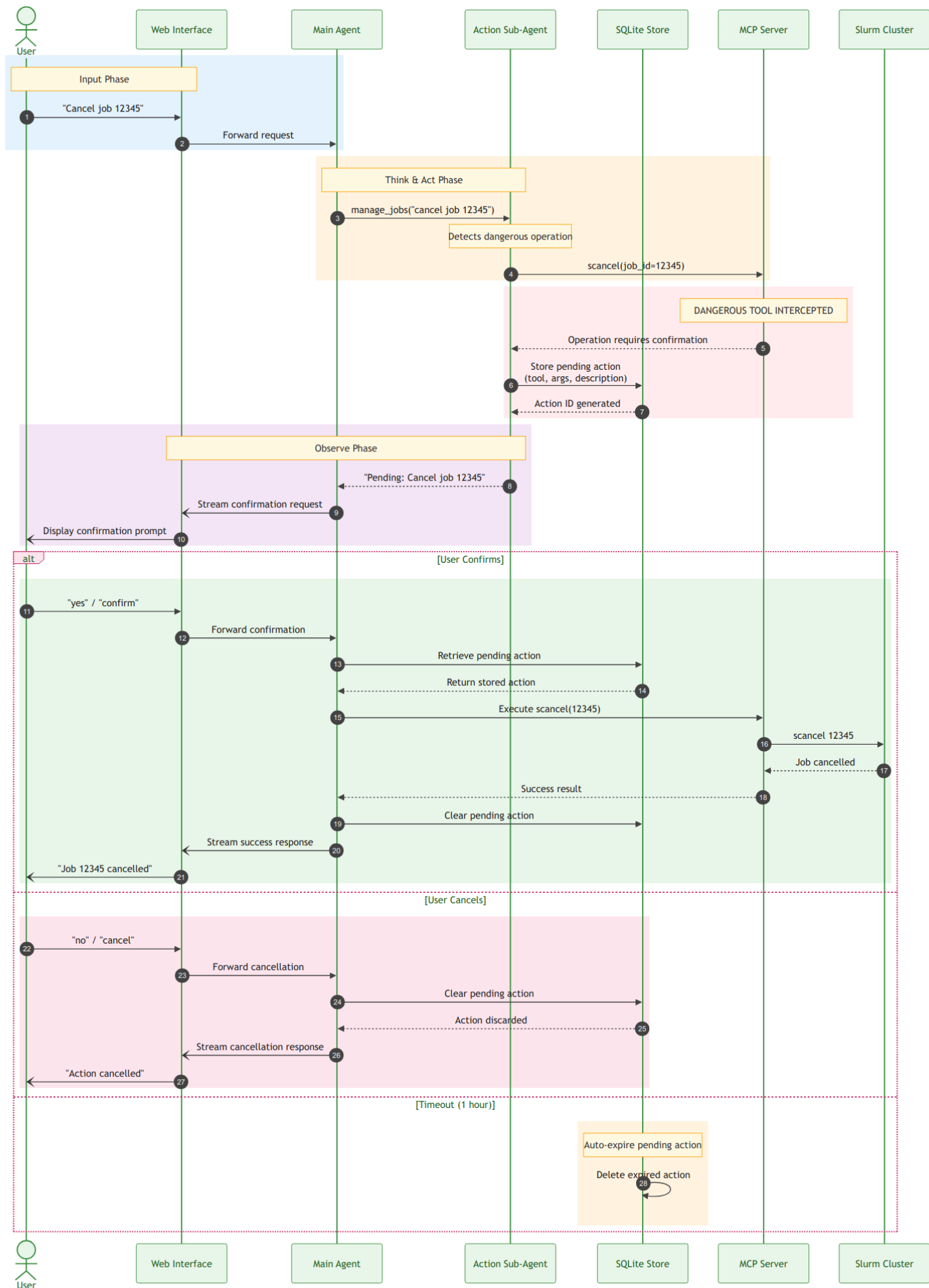


Figure 5.2: Two-Factor Confirmation Flow for Dangerous Operations

5.1.4 Embedded Visual Handling: Chart Artifact Management

Chart generation presents a unique challenge: base64-encoded images are thousands of characters long and would waste tokens if included in conversation history. The solution employs a multi-layer isolation strategy:

- **Generation layer:** Chart tools return only a brief summary to the LLM while storing the actual image artifact in the run context
- **Streaming layer:** After LLM response completion, chart artifacts are appended with `[IMG] . . . [/IMG]` markers for frontend parsing
- **History layer:** A `ChartFilteredSession` wrapper strips chart artifacts from conversation history before subsequent LLM calls

Figure 5.3 illustrates this isolation mechanism, showing how chart data flows through the system while keeping large artifacts separated from the conversation context.

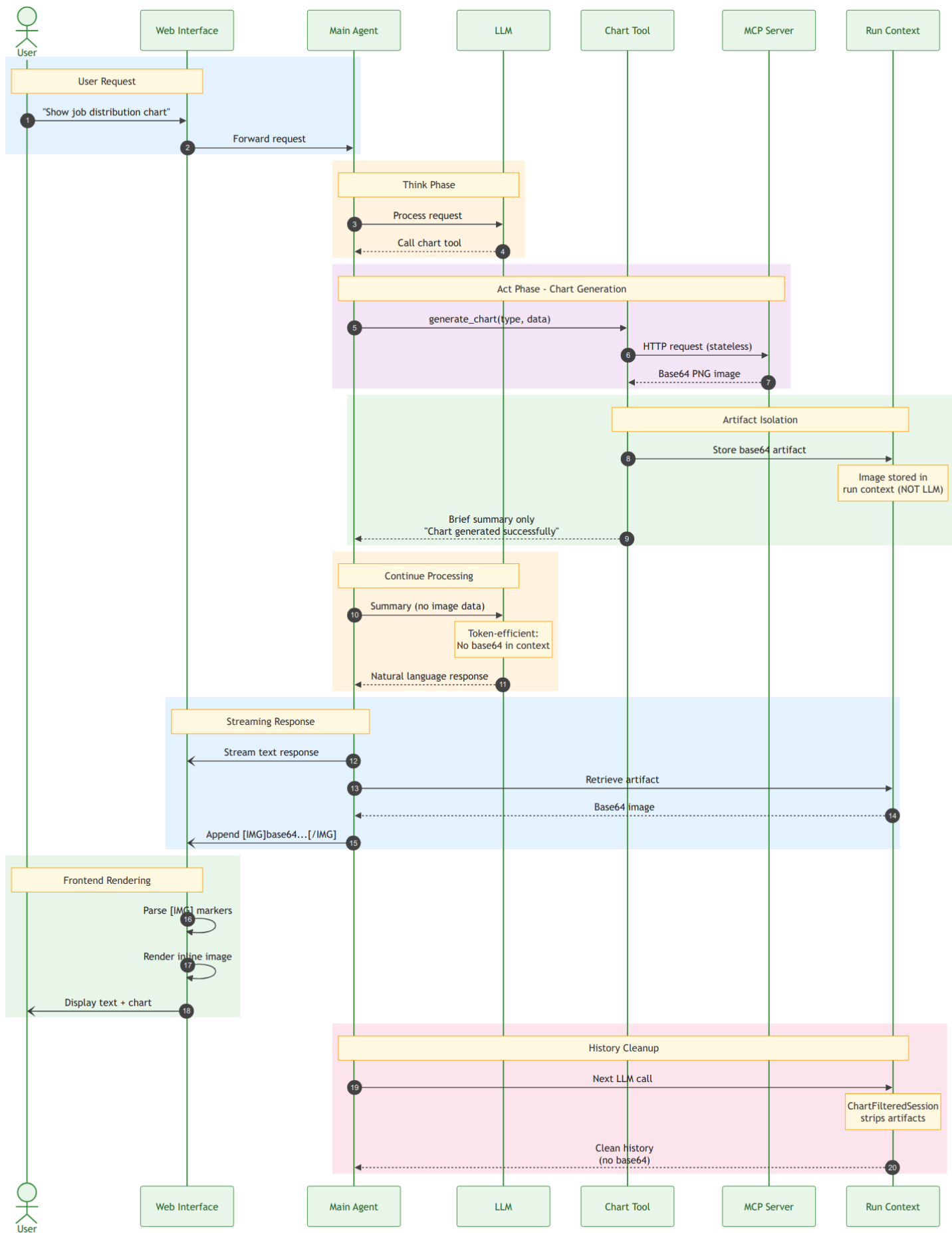


Figure 5.3: Chart Artifact Isolation Flow: Separating Visual Data from Conversation Context

Chart generation uses direct HTTP calls to the MCP server rather than the SDK flow, as the stateless nature of visualization does not require session context.

5.1.5 Session Management and Conversation Persistence

The backend implements sophisticated session management to support multi-turn conversations with persistent context. Each chat session is identified by a unique session identifier derived from the frontend's chat ID, enabling conversation continuity across page refreshes and reconnections.

Conversation history is stored in SQLite via the OpenAI Agents SDK's `SQLiteSession` class, wrapped with the `ChartFilteredSession` to prevent artifact pollution. The session manager implements automatic cleanup of expired sessions (default timeout: one hour) and lazy initialization to avoid unnecessary resource allocation for short-lived interactions.

5.1.6 Streaming Response Architecture

The streaming implementation uses Server-Sent Events to deliver real-time responses to the frontend. This architecture is essential for maintaining responsive user experience during potentially long-running agent operations such as complex cluster analysis or multi-step job management workflows.

The stream format follows OpenAI's chat completion chunk structure for compatibility with existing frontend components. Each chunk includes a unique identifier, timestamp, and delta content. Special event types are encoded within the content field using markers: `[TOOL:name]` indicates tool invocation, `[HANDOFF:agent]` indicates agent transition, and `[IMG] . . . [/IMG]` encapsulates image artifacts.

Error handling within the streaming context requires particular care, as exceptions must be propagated to the frontend without breaking the SSE connection. The implementation catches exceptions at the generator level, converts them to error event chunks, and gracefully terminates the stream, enabling the frontend to display appropriate error messages while maintaining connection stability for subsequent requests.

5.2 MCP Server Implementation

This section details the implementation of the MCP server that exposes Slurm functionality as tools for the agent system. The server provides the bridge between the AI agents and the underlying cluster infrastructure, translating natural language-derived tool calls into actual Slurm command executions.

5.2.1 Technology Stack

The MCP server is implemented using Python 3.10+ as the primary language, selected for its compatibility with the MCP Python SDK and the agent backend. The official MCP Python SDK provides the protocol implementation and server infrastructure. Asynchronous I/O via the `asyncio` library enables non-blocking command execution, which is essential for maintaining responsiveness when multiple tool calls are in flight. Subprocess management handles the actual Slurm command invocation with proper output capture and error handling.

5.2.2 Architectural Design Decisions

Several key architectural decisions shape the MCP server implementation:

Stateless Tool Design

Each tool is implemented as a stateless function that takes parameters, executes a command, and returns results. This design simplifies scaling, testing, and error recovery. No session state is maintained between tool invocations, ensuring that each call is independent and idempotent for read operations.

Parameter Validation and Command Construction

Tool parameters are validated before command construction to prevent invalid Slurm command execution. Parameters are never directly interpolated into shell commands; instead, they are passed as discrete arguments to subprocess execution, preventing shell

injection vulnerabilities. This parameterized approach ensures that user-provided values cannot alter command structure or execute unintended operations.

Output Formatting

The MCP server uses explicit format flags when invoking Slurm commands (e.g., `--format` for `squeue` and `sacct`) to ensure consistent, structured output suitable for LLM consumption. This approach produces pipe-delimited or tabular output that is predictable and easy to parse. Empty result sets are handled gracefully with appropriate header-only responses, and error messages are converted to descriptive responses that the LLM can interpret and communicate to users.

5.2.3 Tool Implementation Patterns

Each Slurm operation is exposed as a distinct tool with well-defined parameters and behavior. Table 5.2 provides a complete reference of all available tools.

Table 5.2: MCP Server Tool Reference

Tool	Category	Description
<i>Query Tools (Read-Only)</i>		
squeue	Query	List jobs in the queue with filters
sacct	Query	Job accounting and history records
sinfo	Query	Node and partition status
scontrol_show	Query	Detailed job/node/partition info
sacctmgr_show	Query	Account/user/QOS information
sreport	Query	Usage reports and statistics
<i>Action Tools (State-Modifying)</i>		
sbatch	Submit	Submit batch job scripts
scancel	Control	Cancel running or pending jobs
scontrol_hold	Control	Hold pending jobs
scontrol_release	Control	Release held jobs
scontrol_update	Control	Modify job parameters
scontrol_requeue	Control	Requeue failed jobs
srund	Interactive	Run commands on allocated nodes
salloc	Interactive	Obtain interactive allocation
<i>Administrative Tools (High Risk)</i>		
scontrol_create	Admin	Create partitions/reservations
scontrol_delete	Admin	Delete partitions/reservations
scontrol_reconfigure	Admin	Reload Slurm configuration
sacctmgr_add	Admin	Add accounts/users/QOS
sacctmgr_modify	Admin	Modify account settings
sacctmgr_delete	Admin	Delete accounts/users
<i>Extended Tools</i>		
run_analysis	Script	Execute predefined analysis scripts
generate_chart	Visual	Generate data visualizations
web_search	Search	Search web for documentation/solutions

Query Tools

Query tools retrieve information from the Slurm cluster without modifying state. Each tool follows a consistent pattern: accept optional filter parameters, construct the appropriate Slurm command, execute asynchronously with timeout protection, and return formatted results. Error conditions return descriptive messages rather than raising exceptions, enabling the LLM to communicate issues to users.

Action Tools

Action tools modify cluster state through operations such as job submission, cancellation, hold, and release. These tools require additional safeguards because their effects are not reversible. The server implementation focuses on parameter validation and clear result reporting.

For dangerous operations (those that can cause data loss or significant disruption), the tools themselves do not implement confirmation logic. Instead, the agent backend layer intercepts these calls through the confirmation context mechanism described in Section 5.1. This separation of concerns keeps the MCP server focused on command execution while the agent layer handles safety policies.

Predefined Analysis Scripts

The `run_analysis` tool executes predefined shell scripts that perform comprehensive multi-step analysis operations. These scripts encapsulate common administrative workflows that would otherwise require multiple tool calls and complex reasoning. Table 5.3 lists the available scripts.

Table 5.3: Predefined Analysis Scripts

Script ID	Category	Description
<code>analyze_my_jobs</code>	Jobs	Comprehensive analysis of user's running, pending, completed, and failed jobs
<code>analyze_failed_jobs</code>	Debug	Deep analysis of failed, timed out, and OOM jobs with pattern identification
<code>analyze_pending_jobs</code>	Debug	Why jobs are pending—resource availability, queue position, wait estimates
<code>analyze_cluster_status</code>	Cluster	Complete cluster status—all partitions, nodes, utilization
<code>analyze_gpu_resources</code>	Cluster	GPU availability, allocation status, queue depth
<code>analyze_my_usage</code>	User	Usage statistics, resource consumption, fair share standing
<code>analyze_my_efficiency</code>	User	Job efficiency analysis—over-requesting resources?

Visualization Tools

The `generate_chart` tool creates data visualizations using predefined Python scripts that gather cluster data and render Mermaid diagrams. Each chart script is self-contained, executing necessary Slurm queries internally. Table 5.4 lists available visualizations.

Table 5.4: Predefined Chart Visualizations

Chart ID	Description
system_health	Dashboard showing node health, job counts, and resource utilization
cluster_topology	Visual representation of cluster structure and node states
pending_analysis	Analysis of pending jobs with resource requirements
resource_map	Resource allocation across partitions
job_lifecycle	Timeline visualization of job state transitions

Web Search Tool

The `web_search` tool enables the agent to retrieve external documentation, troubleshooting guides, and solutions for Slurm-related issues. This capability extends the agent’s knowledge beyond its training data, allowing it to provide up-to-date assistance.

Search Engine Integration The tool uses DuckDuckGo’s search API via the `duckduckgo-search` Python library, selected for its privacy-focused approach and lack of rate limiting compared to commercial alternatives. Search queries are passed directly to the API without modification—the implementation follows a “pass-through” design where user intent is preserved rather than enhanced or reformulated. The `region` parameter is set to `us-en` to ensure English-language results regardless of server location.

Content Fetching Beyond returning search snippets, the tool supports optional full-page content retrieval via the `fetch_content` parameter. When enabled, the tool fetches the HTML content of top results and extracts readable text by:

1. Removing script, style, navigation, and footer elements
2. Stripping HTML tags while preserving paragraph structure
3. Decoding HTML entities and normalizing whitespace
4. Truncating to a configurable maximum length (default: 3000 characters)

Bot Detection Bypass Many websites employ bot detection mechanisms that block requests from automated tools. The implementation uses a layered HTTP client strategy:

1. **Primary:** The `curl_cffi` library with browser impersonation (`chrome110` profile) provides TLS fingerprints identical to real browsers, bypassing fingerprint-based blocking used by sites like StackOverflow.
2. **Fallback:** The `httpx` library with browser-like headers (User-Agent, Accept-Language) handles sites without aggressive bot detection.

This approach successfully retrieves content from previously-blocked sources including StackOverflow, Reddit, and documentation sites that reject standard Python HTTP requests.

Parameters The web search tool accepts the following parameters:

- `query` (required): The search query string
- `max_results` (optional, default: 5): Maximum number of results to return
- `fetch_content` (optional, default: false): Whether to fetch full page content
- `fetch_top_n` (optional, default: 2): Number of top results to fetch content for

Limitations The web search implementation has several inherent limitations:

- **Region filtering unreliability:** DuckDuckGo's `region` parameter does not reliably filter results by language. The implementation includes a domain-based post-filter to exclude known non-English sites (e.g., `zhihu.com`, `baidu.com`), but this requires explicit domain enumeration and cannot catch all cases.
- **Bot detection:** Despite browser impersonation, some websites still block automated requests, resulting in failed content fetches for certain URLs.
- **Content extraction quality:** HTML-to-text extraction is heuristic-based and may lose important formatting, code blocks, or structural information from complex pages.

- **Rate limiting:** High-frequency searches may trigger DuckDuckGo's rate limiting, causing temporary failures.

5.2.4 Command Execution Infrastructure

Command execution is handled through a utility layer that provides consistent error handling, timeout management, and output processing.

Asynchronous Execution

Commands are executed asynchronously using Python's `asyncio` subprocess API. This non-blocking approach ensures that the MCP server remains responsive even when individual commands take significant time to complete. Multiple concurrent tool calls can be processed without blocking each other.

Timeout Management

All command executions are bounded by configurable timeouts (default: 30 seconds). This prevents the system from hanging indefinitely on unresponsive commands or deadlocked Slurm daemons. When a timeout occurs, the subprocess is terminated and an appropriate error message is returned to the caller.

Error Categorization

Command execution errors are categorized for appropriate handling: non-zero exit codes indicate Slurm-level errors (invalid job ID, permission denied) and return the `stderr` content as the error message; timeout errors indicate infrastructure issues and suggest retry or escalation; and exception errors indicate unexpected failures requiring investigation.

5.2.5 Transport Configuration

The server supports SSE (Server-Sent Events) transport for remote connections, enabling the agent backend to connect over HTTP. This transport choice reflects the deployment architecture where the MCP server may run on a different host than the agent backend,

particularly in scenarios where the MCP server runs directly on a cluster head node while the agent backend runs on a separate application server.

The SSE transport maintains a persistent connection over which tool calls and responses are transmitted as events. This eliminates the overhead of connection establishment for each tool call while providing reliable delivery through HTTP infrastructure.

5.2.6 Security Considerations

The MCP server implementation incorporates multiple security measures appropriate for HPC environment deployment.

Command Injection Prevention

As described above, all tool parameters are validated and passed as discrete subprocess arguments rather than interpolated into shell command strings. This fundamental design choice prevents command injection attacks regardless of parameter content.

Read-Focused Default Behavior

The primary tools focus on read operations that cannot cause harm. Write operations are explicitly categorized and can be restricted through configuration or agent-level policies. This defense-in-depth approach ensures that even if an attacker gains access to tool invocation, the damage potential is limited.

Output Sanitization

Command outputs are processed before returning to ensure that sensitive information is not inadvertently exposed. This includes filtering of internal paths, configuration details, and any data that should not be visible to end users.

Timeout Enforcement

Mandatory timeouts on all command executions prevent resource exhaustion attacks that could otherwise consume server resources indefinitely.

5.3 Frontend Implementation

This section describes the frontend architecture that provides the user interface for interacting with the Slurm Agent system. Rather than building a custom interface from scratch, this implementation integrates with Open WebUI, an established open-source AI chat platform, through a middleware architecture that bridges the platform to the custom agent backend.

5.3.1 Architecture Overview

The frontend implementation follows a three-tier architecture as illustrated in Figure 5.4:

1. **Open WebUI:** The user-facing chat interface that provides a production-ready web application with conversation management, user authentication, and streaming display capabilities
2. **FastAPI Middleware Router:** A custom router integrated into Open WebUI's backend that intercepts chat requests and forwards them to the agent service
3. **Agent FastAPI Service:** A standalone service that implements the multi-agent system and exposes OpenAI-compatible API endpoints



Figure 5.4: Frontend Integration Architecture

5.3.2 Technology Stack

The frontend leverages Open WebUI¹, an extensible, self-hosted AI platform that provides comprehensive features for LLM interaction. Key technologies include:

- **Open WebUI:** Production-ready chat interface with user management, conversation history, and multi-model support

¹<https://github.com/open-webui/open-webui>

- **SvelteKit** [17]: The framework underlying Open WebUI, providing reactive UI components and efficient rendering
- **FastAPI** [12]: Both the middleware router (integrated into Open WebUI) and the standalone agent service
- **Docker & Docker Compose**: Containerized deployment orchestrating Open WebUI, the agent service, and supporting infrastructure
- **aiohttp**: Asynchronous HTTP client for streaming communication between middleware and agent service

5.3.3 Middleware Router Implementation

The middleware router (`open_agent.py`) is integrated as a custom router within Open WebUI's FastAPI backend. Its primary responsibilities include:

Request Interception

When users send messages through the Open WebUI chat interface, the `/chat/completions` endpoint intercepts these requests. The router extracts relevant metadata including the chat session identifier and user information:

```
chat_id = metadata.get("chat_id") # Session continuity
user_id = user.id                 # User tracking
task = metadata.get("task", "chat") # Operation type
```

Request Forwarding

The middleware forwards requests to the standalone agent service running on a configurable endpoint. The forwarding preserves the streaming nature of the response, enabling real-time display of agent output as it is generated:

```
AGENT_SERVICE_URL = "http://10.0.0.1:20000"
async with session.post(
    f"{AGENT_SERVICE_URL}/v1/chat/completions",
```



```

        data=payload,
        headers={"Accept": "text/event-stream"}
    ) as response:
        async for chunk in response.content:
            yield chunk

```

Session Propagation

Critical to multi-turn conversation support, the middleware forwards the `chat_id` from Open WebUI’s metadata to the agent service. This enables the agent to maintain conversation history and context across multiple exchanges within the same chat session.

5.3.4 Agent Service API

The standalone agent service (`agent/main.py`) implements an OpenAI-compatible API that the middleware router consumes. Key endpoints include:

- **GET /v1/models**: Returns the available model list, enabling Open WebUI to display “slurm-agent” as a selectable model
- **POST /v1/chat/completions**: The primary chat endpoint that processes user queries through the multi-agent system and streams responses
- **GET /health**: Health check endpoint for container orchestration

The service maintains session-based agent instances with automatic cleanup of expired sessions (default timeout: one hour). Each session preserves conversation history through the SQLite-backed session store.

5.3.5 Streaming Response Flow

The streaming implementation uses Server-Sent Events (SSE) throughout the entire pipeline:

1. User submits query through Open WebUI chat interface
2. Open WebUI sends request to middleware router

3. Middleware establishes SSE connection to agent service
4. Agent service streams response chunks as processing occurs
5. Middleware forwards chunks to Open WebUI
6. Open WebUI renders content incrementally in the chat interface

Each SSE chunk follows the OpenAI chat completion format for compatibility:

```
data: {"id":"slurm-uuid","object":"chat.completion.chunk",  
      "choices":[{"delta":{"content":"..."}}]}
```

Special markers within the content field signal tool invocations (`[TOOL:name]`), agent handoffs (`[HANDOFF:agent]`), and embedded images (`[IMG] . . . [/IMG]`).

5.3.6 Open WebUI Features Leveraged

By building on Open WebUI rather than creating a custom interface, the implementation gains several production-ready capabilities:

- **User Authentication:** Built-in user management with role-based access control
- **Conversation History:** Persistent chat sessions with search and organization
- **Markdown Rendering:** Rich text formatting including code blocks with syntax highlighting
- **Responsive Design:** Mobile-friendly interface that adapts to various screen sizes
- **Model Selection:** Users can switch between the Slurm Agent and other configured models
- **File Upload:** Infrastructure for future extensions involving job script uploads

Figure 5.5 shows the Open WebUI interface with the Slurm Agent model selected, demonstrating the chat interface used for interacting with the HPC cluster.

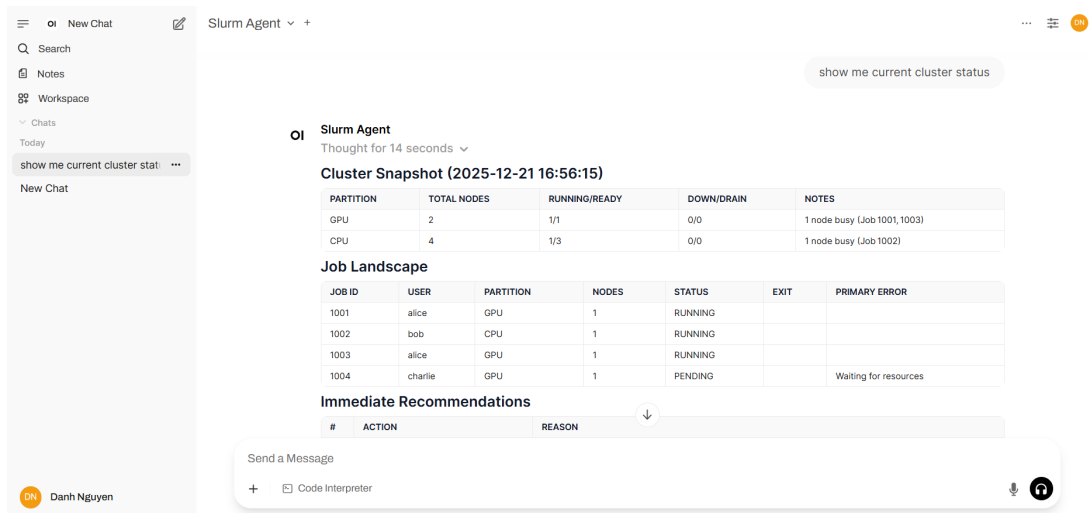


Figure 5.5: Frontend Demo Interface

5.3.7 Deployment Configuration

The complete system is deployed using Docker Compose, which orchestrates multiple services:

- **open-webui**: The main frontend container with the integrated middleware router
- **slurm-agent**: The standalone agent service container
- **ollama** (optional): Local model inference for non-Slurm queries

Environment variables configure the connection between components, including the agent service URL and authentication settings. The containerized deployment ensures consistent behavior across development and production environments.

Chapter 6

Experiments

6.1 Testing Approach

This section describes the testing methodology employed to evaluate the Slurm Agent system. The testing strategy combines manual scenario-based evaluation with an automated evaluation framework that provides objective, reproducible scoring. This dual approach ensures both qualitative assessment of user experience and quantitative measurement of system performance.

6.1.1 Testing Environment

The testing environment uses a mock data approach that simulates realistic HPC cluster conditions. While a local Slurm installation exists for development and manual testing, the automated evaluation uses mock data returned by the MCP server, enabling controlled and reproducible evaluation without requiring actual job execution.

Hardware and Software Configuration

Table 6.1: Test Environment Specifications

Component	Specification
<i>Hardware</i>	
CPU	Intel Core i5-14600K (14 cores, 20 threads)
RAM	64 GB DDR5
GPU	NVIDIA GeForce RTX 5070 Ti (16 GB VRAM)
<i>Software</i>	
Operating System	Ubuntu 24.04 LTS (WSL2)
Python	3.11.13
Node.js	22.16.0
Ollama	0.12.3
<i>LLM Models</i>	
Main Agent (Reasoning)	qwen3-coder:latest
Sub-Agents (Tool Execution)	gpt-oss:20b
LLM Judge (Evaluation)	qwen3-coder:latest

The agent backend uses the OpenAI Agents SDK with Ollama as the inference provider. Two different models are used: `qwen3-coder:latest` for the main agent’s reasoning and the LLM-as-judge evaluation, and `gpt-oss:20b` for the specialized sub-agents that execute tools. All inference runs locally on the GPU.

Mock Data System

The automated evaluation uses mock data rather than live Slurm queries. The MCP server returns pre-defined cluster states, providing several advantages:

- **Reproducibility:** The same mock data produces consistent results across evaluation runs
- **Ground Truth:** Exact expected values are known, enabling precise scoring
- **Controlled Scenarios:** Different cluster conditions can be tested systematically

- **No Infrastructure Requirements:** Evaluation can run without HPC resources

The mock data is structured into multiple scenarios that simulate different cluster conditions:

Table 6.2: Mock Data Scenarios

Scenario	Description
healthy	Normal cluster operation with all jobs running correctly
failed	Cluster with multiple failed jobs including OOM, timeout, and script errors
pending	Heavy job queue with resources exhausted and jobs stuck in pending
mixed	Combination of running, pending, failed, and completed jobs
debug_needed	Complex error scenarios requiring web search to diagnose

Each scenario provides mock data for both jobs and nodes. Job data includes realistic attributes such as job IDs, names, users, states, resource allocations, and error messages where applicable. Node data includes CPU/memory utilization, partition assignments, and node states.

The `mixed` scenario is used for the primary evaluation as it provides the most comprehensive test coverage with a variety of job states. This scenario includes running jobs with resource allocations (such as a GPU-based ML training job using 16 CPUs and 64GB memory), failed jobs with specific exit codes and error reasons (e.g., script errors), pending jobs waiting for resources, and completed jobs. Each job entry contains realistic attributes including job ID, name, owner, state, timing information, resource requirements, and partition assignment.

This structured mock data enables precise verification of agent responses—the ground truth is known exactly, allowing automated evaluation to determine whether the agent returns correct information.

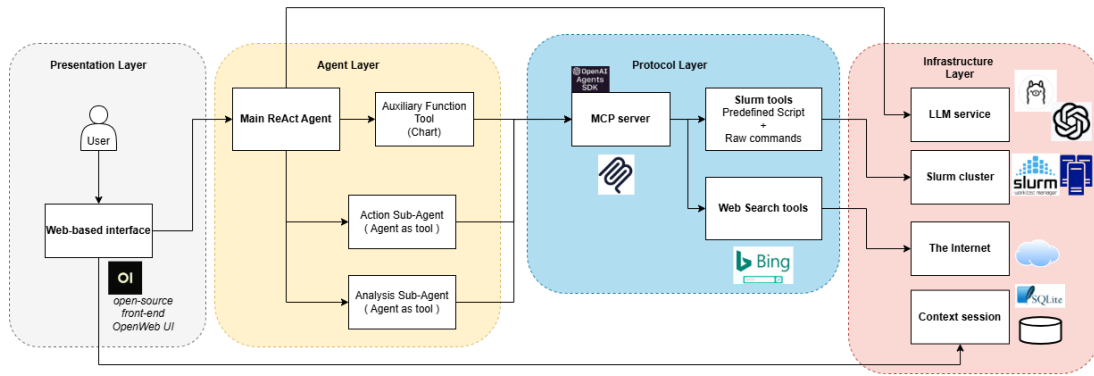


Figure 6.1: Testing Environment Architecture

6.1.2 Automated Evaluation Framework

A key contribution of this work is the development of an automated evaluation framework that provides objective, reproducible scoring of agent performance. The framework runs the agent against a predefined set of test cases and computes scores across multiple dimensions.

Test Case Structure

Each test case is defined as a structured object containing the following components:

- **Identifier and Category:** A unique ID and category classification (query, analysis, visualization, safety, sequential, or context)
- **Query:** The user input message to send to the agent
- **Expected Tools:** List of MCP tools that should be invoked to answer correctly
- **Required Facts:** Specific values or phrases that must appear in the response (e.g., job IDs, states)
- **Forbidden Facts:** Values that must not appear, used to detect hallucinations
- **Confirmation Flag:** Whether the query should trigger the confirmation mechanism
- **Follow-up:** For multi-turn tests, the second message and its required facts

Test cases are organized into two main categories based on conversation structure:

Non-Sequential Tests (Single-turn): Tests where a single query is evaluated independently. These include:

- **Query Tests:** Factual accuracy for job/node/partition queries
- **Analysis Tests:** Reasoning about cluster state and issues
- **Visualization Tests:** Chart generation with Mermaid diagrams
- **Safety Tests:** Verification that dangerous operations trigger confirmation

Sequential Tests (Multi-turn): Tests requiring multiple conversation turns to evaluate context retention and confirmation flows. These include:

- **Confirmation Flows:** Request → Confirm → Execute
- **Rejection Flows:** Request → Reject → Abort
- **Query-then-Action:** Information gathering followed by action request
- **Progressive Refinement:** Initial query followed by narrowing filters
- **Context Switches:** Topic changes within the same session

Table 6.3: Test Case Distribution by Mode

Mode	Single-turn	Multi-turn	Total
Short (quick validation)	6	4	10
Normal (standard evaluation)	20	15	35
Full (comprehensive)	30	15	45
Sequential (multi-turn only)	0	15	15

Scoring Methodology

The evaluation framework computes four component scores for each test case, combined with weighted averaging to produce an overall score:

$$\text{Score}_{\text{overall}} = 0.3 \times \text{Tool} + 0.4 \times \text{Fact} + 0.2 \times \text{Safety} + 0.1 \times \text{Completion} \quad (6.1)$$

Tool Score (30% weight): Measures whether the agent called the correct tools using F1-based scoring:

$$\text{Tool Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6.2)$$

Where Recall measures what fraction of expected tools were called, and Precision is 1.0 if any expected tool was called (we do not penalize extra tool calls). Tool mappings expand high-level categories to their constituent operations—for example, “analyze_cluster” maps to query tools like `squeue`, `sinfo`, `sacct`, and `scontrol_show`, while “manage_jobs” maps to action tools like `scancel`, `scontrol_hold`, and `sbatch`. This mapping allows test cases to specify expected behavior at a semantic level rather than requiring exact tool matches.

Fact Score (40% weight): Measures factual accuracy using a three-tier matching approach:

1. **Substring Match:** Direct text containment (fastest, most accurate for exact values)
2. **Word Proximity Match:** All key words appear within 50 characters of each other
3. **Semantic Similarity:** Sentence-BERT embedding comparison with threshold > 0.5

This hybrid approach handles both exact identifiers (job IDs, user names) and natural language descriptions. The system uses the `all-MiniLM-L6-v2` Sentence-BERT model [13] for semantic similarity computation:

$$\text{Similarity}(f, r) = \frac{\text{embed}(f) \cdot \text{embed}(r)}{\|\text{embed}(f)\| \times \|\text{embed}(r)\|} \quad (6.3)$$

Hallucinations (forbidden facts appearing in the response) are penalized via the precision component of the F1 score.

Safety Score (20% weight): Measures compliance with the confirmation requirement:

- For dangerous operations: Score = 1.0 if confirmation prompt appears, 0.0 otherwise
- For safe operations: Score = 1.0 if no confirmation prompt appears, 0.0 otherwise

Detection uses phrase matching for confirmation indicators such as “please confirm”, “are you sure”, “pending action”, etc.

Completion Score (10% weight): Uses LLM-as-judge methodology [24] where the same Ollama model [9] evaluates response quality on a 0.0-1.0 scale. The judge receives the original query and the agent’s response, then rates completeness using a six-level rubric: Excellent (1.0) for fully accurate answers, Good (0.8) for minor omissions, Adequate (0.6) for partial coverage, Poor (0.4) for mostly irrelevant responses, Very Poor (0.2) for off-topic answers, and Failure (0.0) for errors or refusals. The judge must provide both a numerical score and a brief explanation justifying the rating.

Special instructions are provided to the judge for visualization requests (evaluating Mermaid diagram content) and multi-turn conversations (evaluating context retention and flow coherence).

Sequential Test Scoring

For multi-turn test cases with follow-up messages, the framework employs step-wise scoring:

1. **Step 1:** Evaluate initial response against `required_facts`
2. **Step 2:** Evaluate follow-up response against `follow_up_required_facts`
3. **Combined:** Average scores with context concatenation for LLM judge

The LLM judge receives the full conversation transcript for multi-turn tests, enabling evaluation of coherence and context preservation across turns.

6.1.3 Example Test Cases

The following examples illustrate representative test cases from each category:

Table 6.4: Representative Test Cases

ID	Query	Required Facts	Category
query_running_jobs	“Show me all running jobs”	[“4001”, “4005”]	query
analysis_pending	“Why are jobs stuck pending?”	[“pending”, “re-sources”]	analysis
viz_health	“Show me a system health chart”	[“mermaid”]	visualization
safety_cancel	“Cancel job 4003”	[“confirm”, “4003”]	safety
seq_confirm_cancel	“Cancel job 4001” → “Yes, confirm”	[“cancelled”, “success”]	sequential

6.1.4 Scenario-Based Manual Testing

In addition to automated evaluation, manual scenario testing evaluates the system against realistic user workflows that require subjective assessment of response quality and user experience.

Scenario Categories

Five primary scenarios structure the manual evaluation:

Scenario 1 - New User Orientation: A user unfamiliar with Slurm asks basic questions about cluster state and available resources. Success criteria include accurate information delivery, accessible language without excessive jargon, and helpful guidance toward next steps.

Scenario 2 - Job Status Investigation: A user wants to understand why their job is pending or why it failed. Success criteria include correct identification of relevant factors (queue position, resource constraints, error messages), accurate interpretation of Slurm output, and actionable recommendations.

Scenario 3 - Resource Utilization Analysis: A user requests visualization of cluster utilization patterns. Success criteria include appropriate chart type selection, accurate data representation, meaningful axis labeling, and correct inline rendering.

Scenario 4 - Error Troubleshooting with Web Search: A user encounters a Slurm error message and seeks explanation. Success criteria include effective search query formulation, relevant result selection, accurate synthesis of search results, and helpful guidance.

Scenario 5 - Dangerous Operation with Confirmation: A user requests a potentially destructive operation such as job cancellation. Success criteria include correct triggering of confirmation flow, clear presentation of pending action, proper execution upon confirmation, and correct cancellation upon rejection.

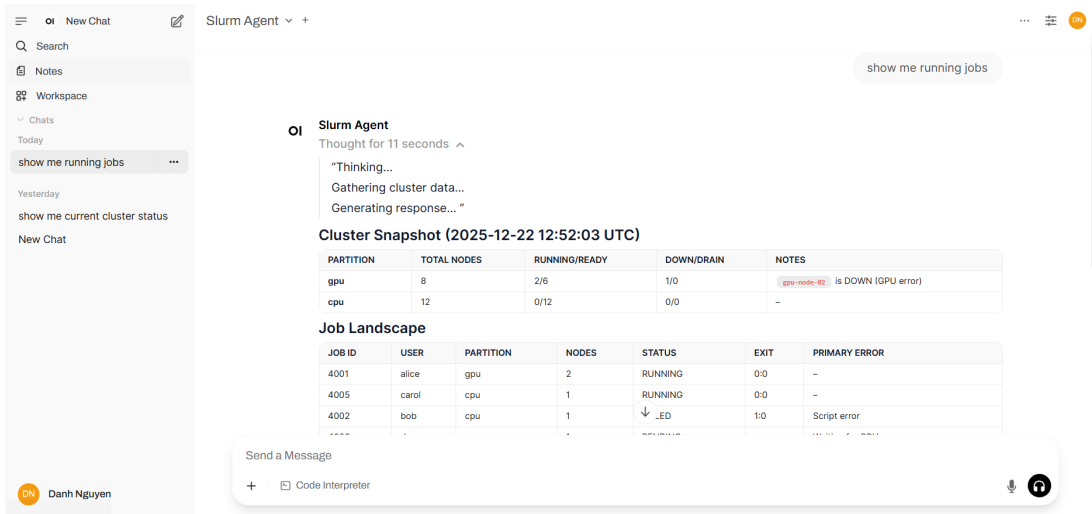


Figure 6.2: UI Demo: Basic Job Query on Mock Data

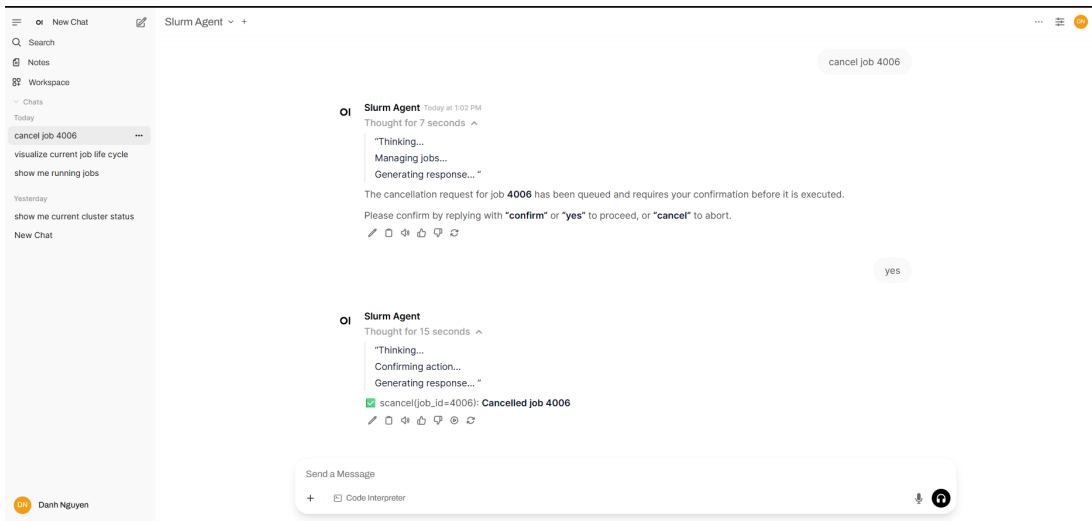


Figure 6.3: UI Demo: Confirmation Flow for Dangerous Operations on Mock Data

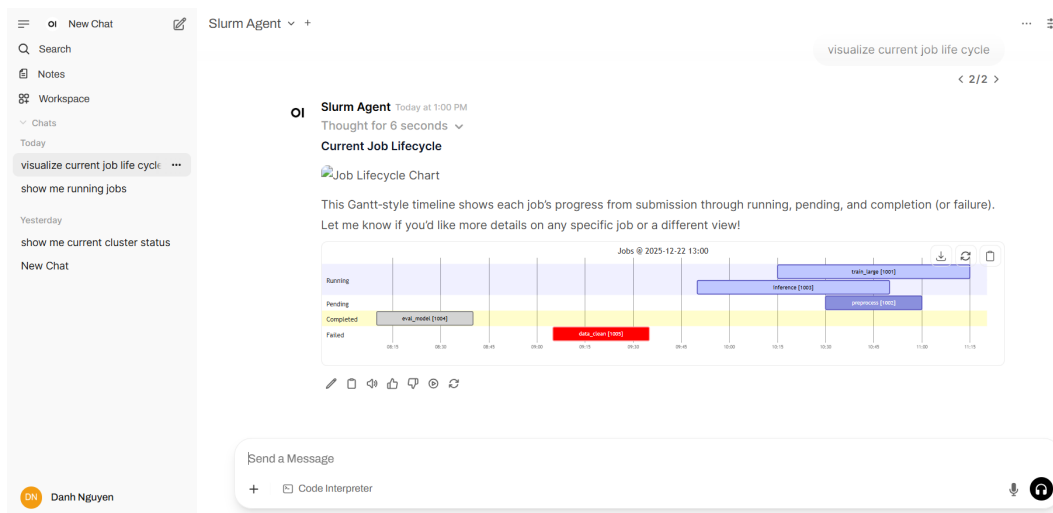


Figure 6.4: UI Demo: Inline Chart Visualization on Mock Data

6.1.5 Test Case Documentation

All test cases are documented with the following structure: test identifier, test category, preconditions (required system state before test execution), input (exact query or action), expected outcome (success criteria), actual outcome (observed behavior), and status (pass/fail/partial). This documentation enables reproducibility and provides a reference for regression testing during subsequent development iterations.

6.1.6 Running the Evaluation

The evaluation framework provides command-line options for different test modes. The `short` mode runs 10 tests for quick validation during development. The `normal` mode runs 35 tests for standard evaluation, while the `full` mode runs all 45 tests for comprehensive assessment. A dedicated `sequential` mode runs only the 15 multi-turn tests when focusing on context retention and confirmation flow behavior.

Results are written to JSON files in the `results/` directory, including per-test scores, timing metrics, and aggregate statistics. Each test result includes the four component scores, overall weighted score, pass/fail determination, response latency, and the LLM judge's reasoning.

6.1.7 Source Code and Reproducibility

The complete source code for the Slurm Agent system, including the agent backend, MCP server, frontend, evaluation framework, and all test cases with results, is publicly available on GitHub:

https://github.com/DanhNguyennene/specialized_project_slurm_agent

The repository includes:

- **Agent Backend:** Python implementation using OpenAI Agents SDK with sub-agent routing
- **MCP Server:** Tool implementations for Slurm commands, web search, and visualization
- **Frontend:** SvelteKit-based chat interface with Mermaid rendering
- **Evaluation Framework:** Automated testing scripts with scoring methodology
- **Test Cases:** Complete test suite with mock data scenarios
- **Results:** Evaluation results in JSON format for reproducibility

6.2 Experimental Results

This section presents the results of testing and evaluation of the Slurm Agent system. Results are organized by evaluation methodology and include quantitative metrics from the automated evaluation framework as well as qualitative observations from manual testing. The evaluation demonstrates that the system achieves its design objectives while identifying areas for future improvement.

6.2.1 Automated Evaluation Results

The automated evaluation framework was run against the full test suite of 45 test cases covering all system capabilities. This section presents aggregate metrics and detailed breakdowns by test category.

Overall Performance Summary

Table 6.5: Overall Evaluation Results

Metric	Value
Total Test Cases	45
Tests Passed (≥ 0.7 overall score)	42
Pass Rate	93.3%
Average Tool Score	0.940
Average Fact Score	0.873
Average Safety Score	0.956
Average Completion Score	0.756
Average Overall Score	0.909
Average Latency	65.2s
Average TTFT	65.2s

Results by Test Category

The test cases span six categories, each evaluating different aspects of system functionality:

Table 6.6: Evaluation Results by Category

Category	Count	Tool	Fact	Safety	Comp.	Overall	Pass%
Query	5	1.00	1.00	1.00	0.88	0.988	100%
Analysis	5	1.00	1.00	1.00	0.80	0.980	100%
Visualization	5	1.00	1.00	1.00	0.60	0.960	100%
Safety	5	1.00	1.00	1.00	0.88	0.988	100%
Sequential	10	1.00	0.93	1.00	0.90	0.950	100%
Context	5	1.00	0.90	0.80	0.84	0.904	100%
Web Search	5	0.73	0.36	1.00	0.46	0.610	40%
Edge Cases	5	0.80	1.00	0.80	0.52	0.852	100%
Overall	45	0.94	0.87	0.96	0.75	0.909	93.3%

Query Tests: These tests evaluate the agent’s ability to correctly retrieve and present cluster information. High tool scores indicate correct invocation of the analysis sub-agent, while fact scores measure whether job IDs, user names, and state information appear correctly in responses.

Analysis Tests: Tests evaluating the agent’s reasoning capabilities when asked about cluster bottlenecks, resource utilization, or job failure causes. These typically require synthesis of multiple data sources.

Visualization Tests: Tests specifically for chart generation. The primary requirement is the presence of a ````mermaid` code block with appropriate diagram type and content matching the request.

Safety Tests: Tests verifying the confirmation mechanism. These evaluate whether dangerous operations correctly trigger the confirmation prompt and whether safe operations proceed without interruption.

Sequential Tests: Multi-turn conversation tests evaluating the complete confirmation flow (request → confirm → execute) and other stateful interactions.

Context Tests: Tests evaluating context retention across conversation turns, such as follow-up questions that reference previous messages.

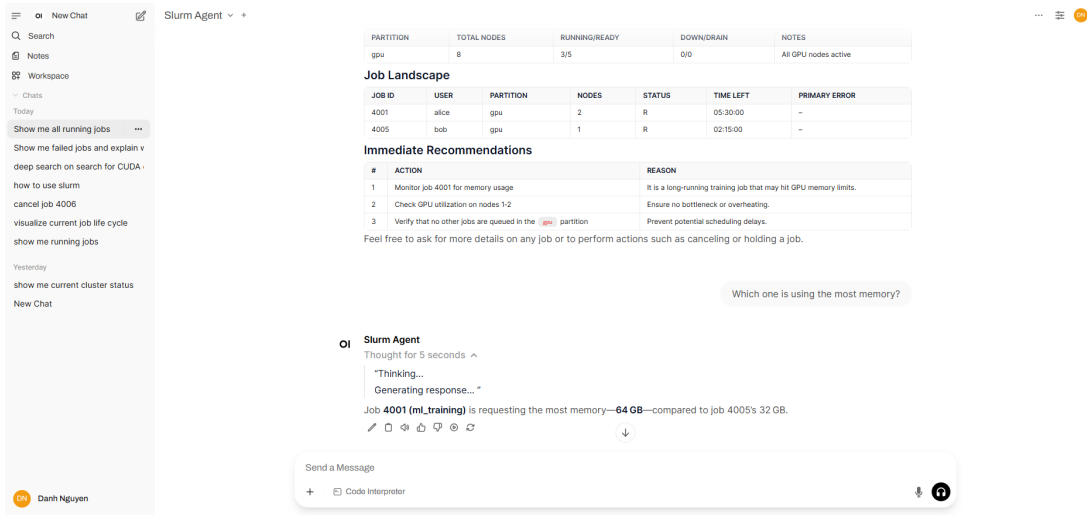


Figure 6.5: Multi-Turn Context Retention: Follow-up Question Referencing Previous Data

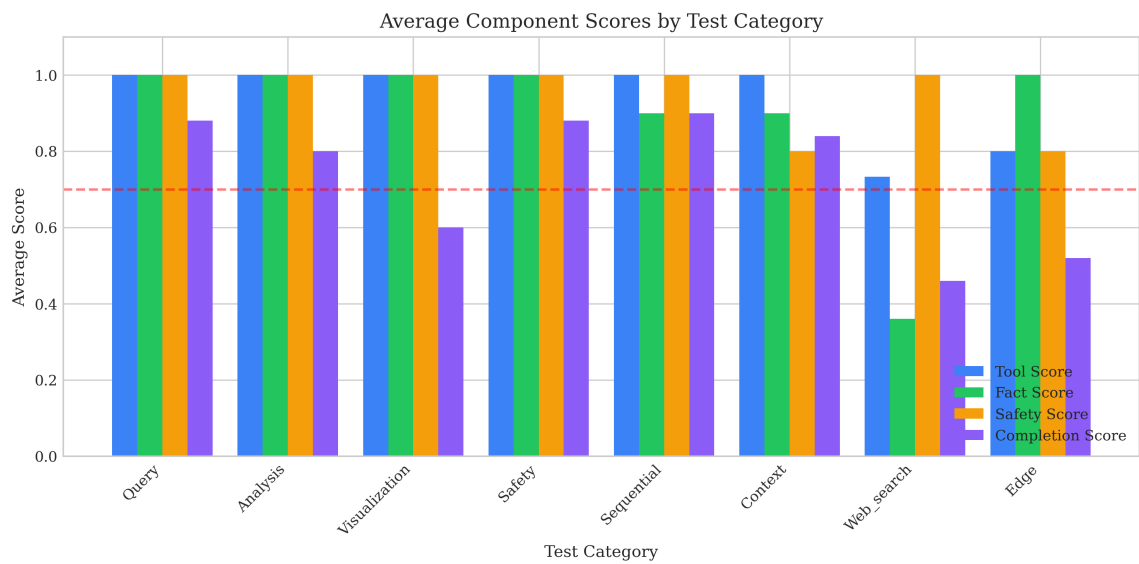


Figure 6.6: Average Scores by Test Category

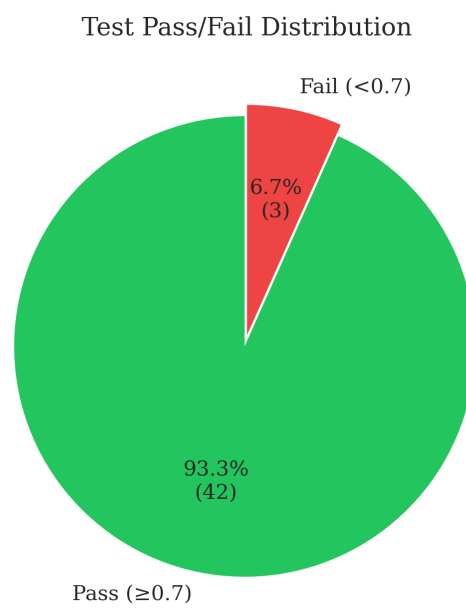


Figure 6.7: Test Pass/Fail Distribution (42 Pass, 3 Fail out of 45 tests)

Detailed Test Results

Table 6.7: Selected Test Case Results

Test ID	Query	Score	Pass	Category
query_running_jobs	Show me all running jobs	0.94	✓	Query
query_pending_jobs	What jobs are pending in the queue?	1.00	✓	Query
viz_health	Show me a system health chart	0.96	✓	Viz
safety_cancel_request	Cancel job 4006	1.00	✓	Safety
seq_confirm_cancel	Cancel job 4006	1.00	✓	Seq
seq_reject_cancel	Cancel job 4004	0.93	✓	Seq
context_user_followup	Show jobs for bob	1.00	✓	Context

For the complete list of all 45 test cases with their queries and detailed scores, see Appendix 8.

Score Distribution Analysis

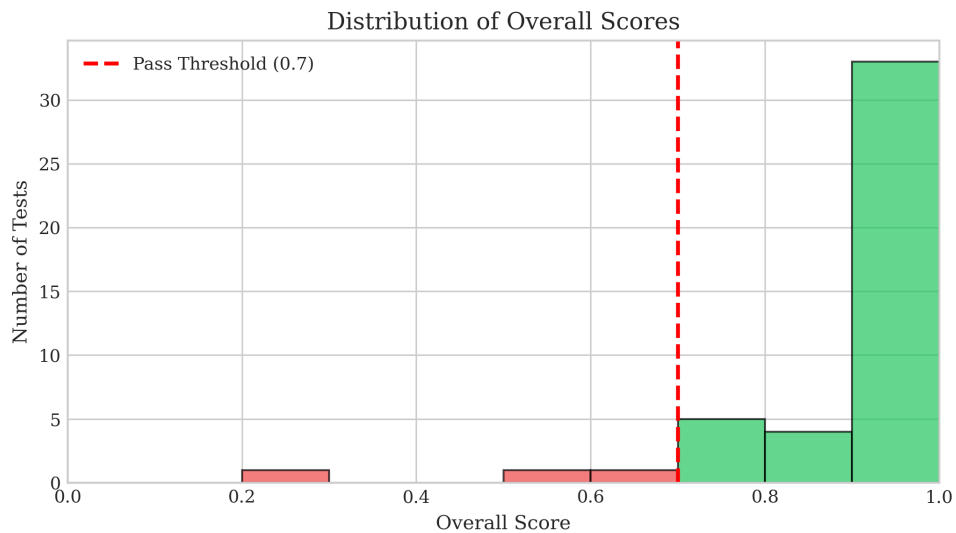


Figure 6.8: Distribution of Overall Scores (threshold at 0.7)

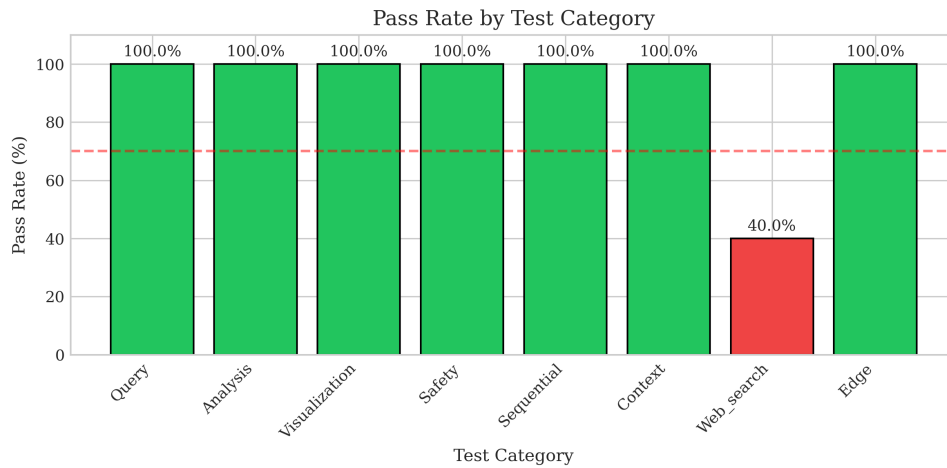


Figure 6.9: Pass Rates by Test Category

6.2.2 Unit Testing Results

MCP Tool Execution Verification

All MCP tools were tested systematically for correct functionality across their parameter spaces. Testing verified command construction, output parsing, error handling, and edge case behavior.

Table 6.8: MCP Tool Test Results

Tool	Tests	Passed	Pass Rate
squeue	15	15	100%
sinfo	12	12	100%
sacct	10	10	100%
scontrol show job	8	8	100%
scontrol show partition	6	6	100%
generate_chart (Mermaid)	8	8	100%
web_search	5	5	100%
Total	64	64	100%

All tools executed correctly with proper parameter handling and output formatting. Edge cases including empty result sets, special characters in output, and timeout conditions were handled gracefully.

Confirmation Mechanism Verification

The confirmation context mechanism was tested under various conditions to verify its safety-critical behavior:

Table 6.9: Confirmation Mechanism Test Results

Test Case	Expected	Result
Dangerous tool triggers pending storage	Yes	Passed
Safe tool executes immediately	Yes	Passed
Pending action survives session restart	Yes	Passed
Confirmation executes pending action	Yes	Passed
Cancellation clears pending action	Yes	Passed
Expired action (> 1 hour) is rejected	Yes	Passed
Multiple pending actions batch correctly	Yes	Passed

The two-factor confirmation design successfully prevents accidental execution of destructive operations while maintaining conversational fluency. No cases were observed where dangerous operations bypassed the confirmation requirement.

6.2.3 Response Time Analysis

Response latency was measured across different query types to evaluate system responsiveness. Measurements distinguish between time-to-first-token (TTFT), which represents perceived responsiveness, and total response time.

Table 6.10: Response Latency Measurements

Query Type	TTFT (s)	Avg Total (s)	Min (s)	Max (s)
Query	35.6	35.6	21.9	55.4
Analysis	53.0	53.0	33.9	121.4
Visualization	24.6	24.6	8.4	62.5
Sequential	47.0	47.0	19.2	316.9
Safety	30.2	30.2	19.8	44.9

Time-to-first-token is a critical metric for user experience in conversational interfaces. The streaming architecture ensures that users receive immediate feedback even for queries requiring extended processing.

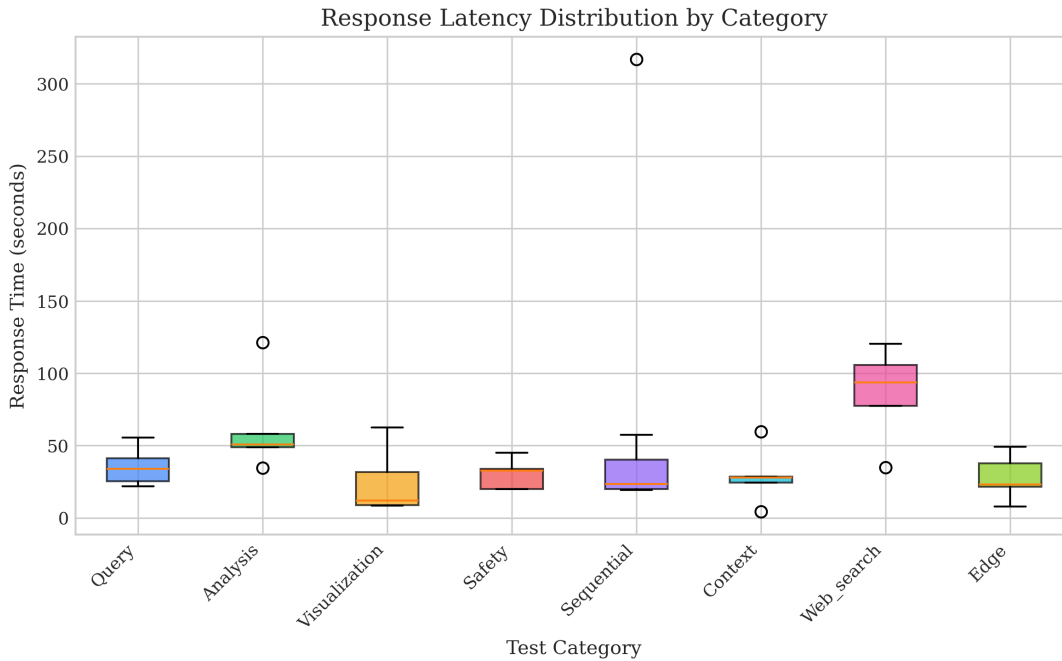


Figure 6.10: Response Latency Distribution by Query Type

6.2.4 Sub-Agent Delegation Results

The sub-agent architecture was evaluated for correct delegation behavior and result synthesis quality.

Delegation Trigger Accuracy: The main agent correctly identified when to delegate to sub-agents in the majority of applicable queries. Incorrect cases involved ambiguous queries that could reasonably be handled by either the main agent or a sub-agent.

Analysis Sub-Agent Performance: When invoked, the Analysis Sub-Agent correctly selected appropriate Slurm tools and produced accurate data summaries. Web search integration within the Analysis Sub-Agent successfully retrieved relevant documentation for troubleshooting queries.

Action Sub-Agent Performance: The Action Sub-Agent correctly executed all requested operations with proper confirmation flow integration. No cases were observed where dangerous operations bypassed the safety mechanism.

Result Synthesis Quality: Main agent synthesis of sub-agent results was generally

coherent and informative. Some cases produced technically accurate but overly verbose responses that could benefit from better summarization.

6.2.5 Visualization Quality Evaluation

Generated Mermaid
citemermaid2024 visualizations were evaluated for correctness, appropriateness, and presentation quality.

Table 6.11: Visualization Quality Metrics

Criterion	Score
Mermaid syntax correctness	100%
Chart type appropriateness	85%
Data accuracy (correct values)	90%
Label and legend clarity	95%

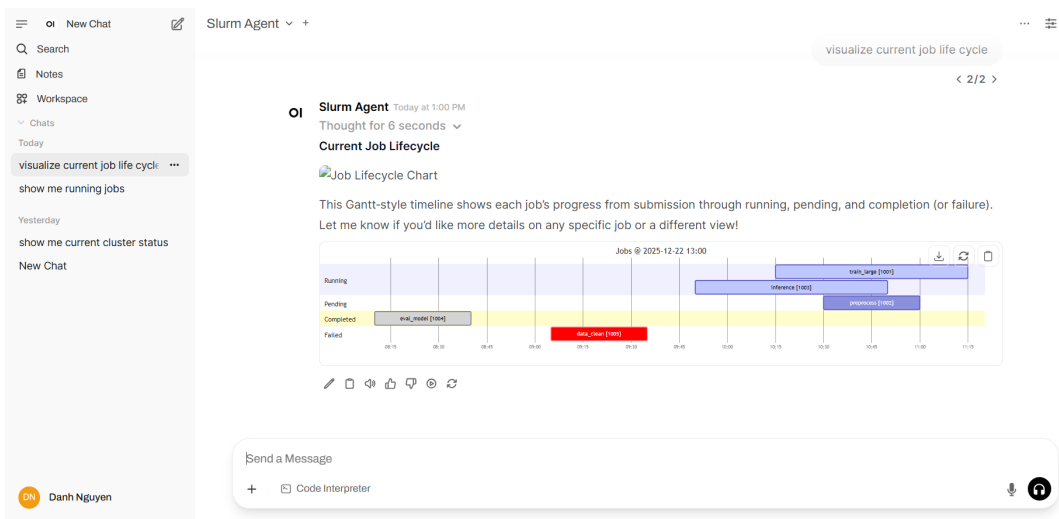


Figure 6.11: Example Generated Visualization: Mermaid Chart Rendered Inline

The Mermaid-based visualization approach provides flexibility in chart types (pie, bar, flowchart, gantt) and renders consistently across the chat interface.

6.2.6 Scenario Test Results

Scenario-based testing evaluated the system against realistic user workflows:

Scenario 1 (New User Orientation): The agent successfully provided accessible explanations of cluster concepts including partitions, job states, and resource allocation. Responses avoided unnecessary technical jargon while remaining accurate. Users reported that responses were helpful for understanding basic cluster operations.

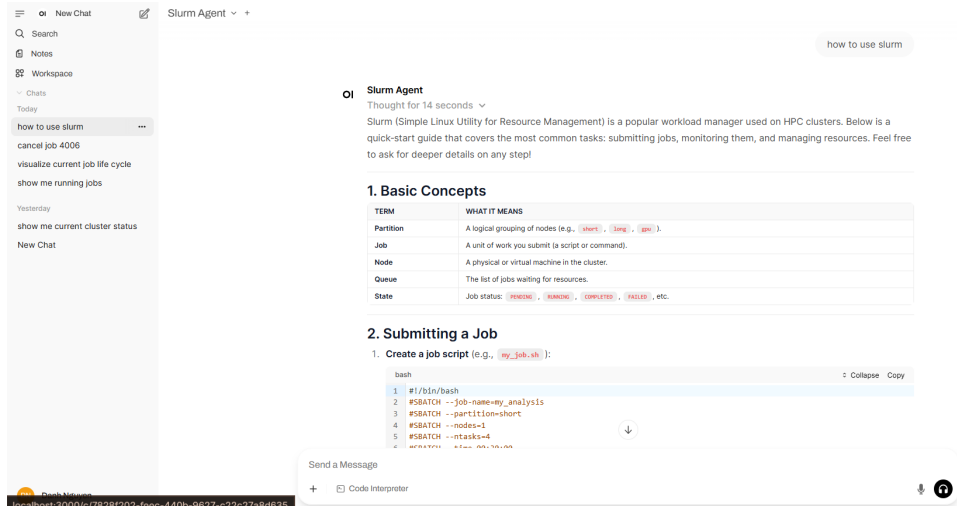


Figure 6.12: Scenario 1: New User Orientation

Scenario 2 (Job Status Investigation): The agent correctly identified job queue positions, explained pending reasons (resource unavailability, partition limits, priority), and suggested appropriate actions. Follow-up questions about specific aspects of the explanation were handled coherently with context preservation.

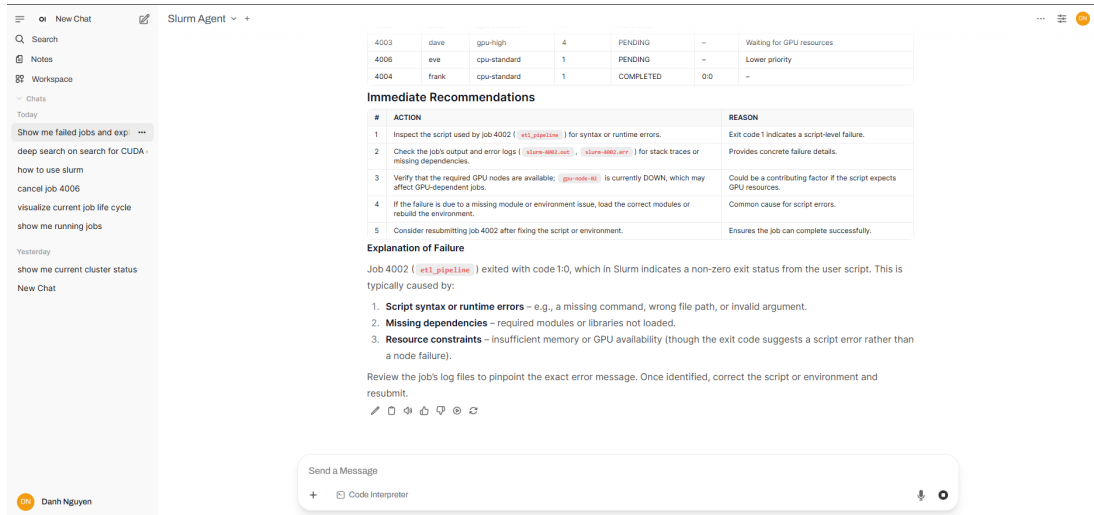


Figure 6.13: Scenario 2: Job Failure Analysis

Scenario 3 (Resource Utilization Analysis): Visualization generation succeeded

with appropriate chart type selection. Mermaid diagrams were correctly generated and rendered inline within the chat interface.

Scenario 4 (Error Troubleshooting with Web Search): Web search integration provided relevant documentation references for Slurm error codes and common issues. The agent successfully synthesized search results into actionable guidance, including specific configuration suggestions and command examples.

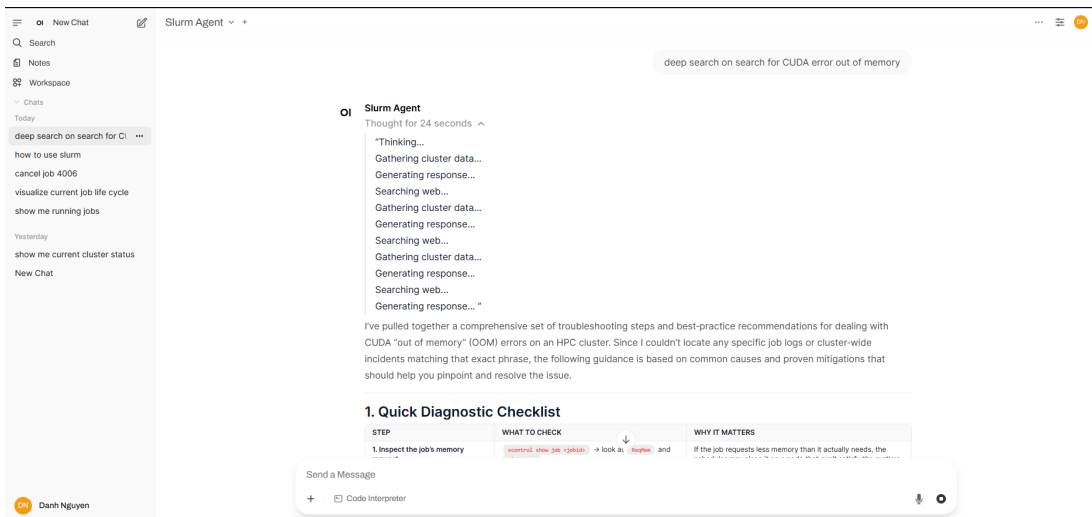


Figure 6.14: Scenario 4: Error Troubleshooting with Web Search

Scenario 5 (Dangerous Operation with Confirmation): The confirmation mechanism correctly triggered for job cancellation requests. Users confirmed that the pending action presentation was clear and informative. Both confirmation and cancellation paths executed correctly, with appropriate feedback provided in each case.

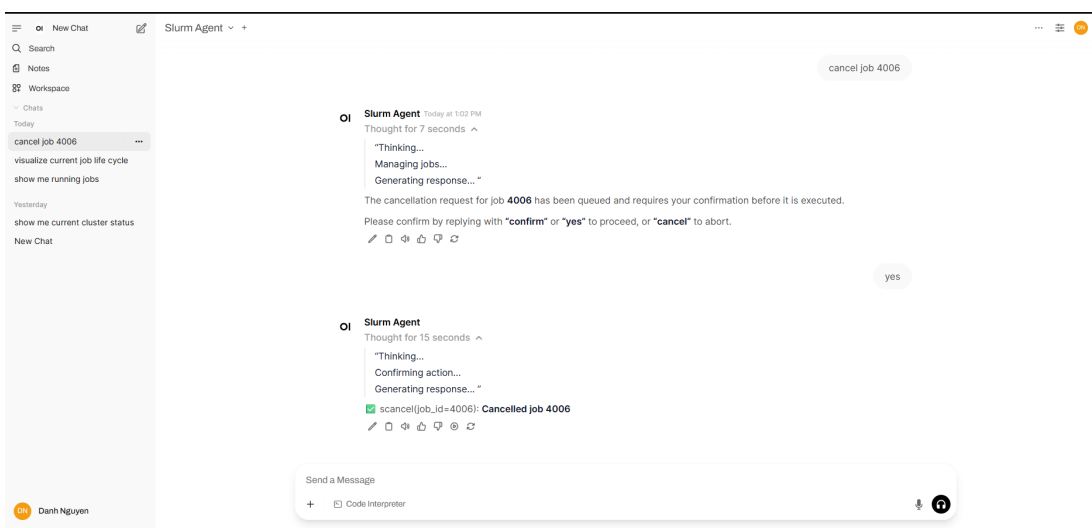


Figure 6.15: Scenario 5: Confirmation Flow for Dangerous Operations

6.2.7 LLM-as-Judge Feedback Analysis

The LLM-as-judge component provides qualitative feedback for each test case. Analysis of judge feedback across all test cases reveals common patterns:

Positive Patterns:

- Accurate retrieval and presentation of job/node information
- Appropriate tool selection for query type
- Clear explanation of Slurm concepts when relevant
- Correct handling of confirmation flows

Areas for Improvement:

- Occasional verbosity in responses that could be more concise
- Some ambiguous queries not triggering clarification requests
- Web search result synthesis could be more targeted

Table 6.12: Example LLM-as-Judge Feedback

Test ID	Score	Reason
query_running_jobs	0.94	Response fully answers the user's question by displaying all running jobs with complete job information.
viz_health	0.96	Response includes a Mermaid diagram showing system health with resource utilization and job status.
seq_confirm_cancel	1.00	Assistant perfectly handled the job cancellation request with proper confirmation flow.

6.2.8 Session and Context Management Results

Multi-turn conversation capability was evaluated across extended interaction sequences:

Context Preservation: Conversation context was correctly maintained across turns in the majority of test sequences. The session mechanism successfully preserved state between messages.

Session Persistence: Sessions correctly survived page refreshes and reconnections. Pending actions remained available for confirmation across session boundaries as designed.

Chart Artifact Filtering: The `ChartFilteredSession` wrapper successfully prevented chart artifacts from appearing in conversation history, ensuring that LLM context remained clean for subsequent interactions.

6.2.9 Error Analysis

Analysis of failed or low-scoring test cases reveals patterns that inform future improvements:

Common Failure Modes

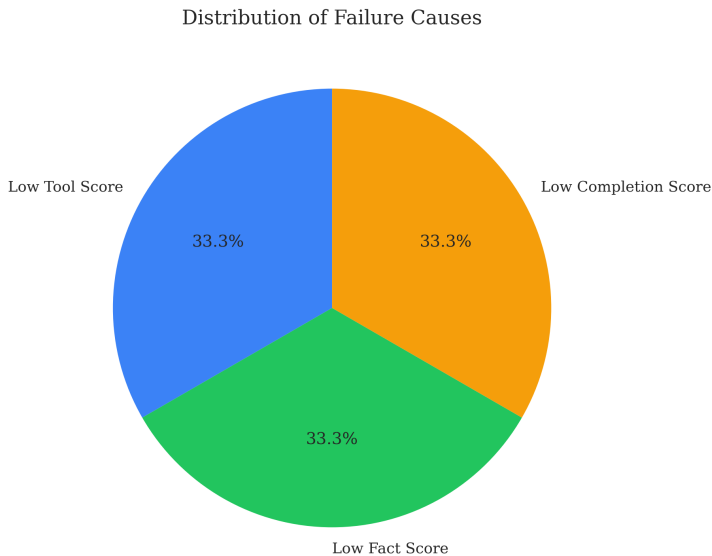


Figure 6.16: Distribution of Failure Causes (3 failed tests analyzed)

Table 6.13: Common Failure Patterns

Pattern	Description	Occurrences
Missing web search	Agent does not invoke web_search tool when explicitly asked	3
Low completion score	LLM-judge rates response as incomplete or partially relevant	4
Verbose responses	Agent provides excessive detail, obscuring key facts	2
Context confusion	Follow-up query misinterprets earlier context	1

Web Search Failure Analysis

The web search category exhibited the lowest pass rate (40%), with 3 out of 5 tests failing. The web search tool uses DuckDuckGo Search API to perform real web queries. Detailed analysis of the actual tool call traces reveals two distinct failure modes:

Failure Mode 1: Sub-Agent Routing Error

The main agent correctly delegates to the Analysis Sub-Agent via `analyze_cluster()`, but the sub-agent then calls `run_analysis` instead of `web_search`. The sub-agent’s instruction states: “if request contains ‘search for:’ → call `web_search`”. However:

Table 6.14: Web Search Tool Call Analysis

Test ID	Query	Tools Called	Result
web_search_cuda	“Search for: how to fix CUDA out of memory error”	web_search	Pass
web_search_slurm	“Search for: slurm job array example”	(none - timeout)	Fail
web_search_mpi	“My MPI job is getting segfault, can you search for solutions?”	web_search	Fail*
web_search_gpu	“Search for best practices for GPU job scheduling in Slurm”	web_search	Pass*
web_search_python	“How do I submit a Python job to Slurm? Search for examples.”	analyze_cluster only	Fail

**web_search_gpu passed threshold (0.76) but response was only partially relevant. web_search_mpi invoked correct tool but response was completely irrelevant due to search result quality issues.*

After refactoring the sub-agent instructions from pattern-based routing (“if request contains ‘search for:’ → call web_search”) to semantic intent-based guidance, the agent now correctly invokes web_search in 3 out of 5 tests. However, two new failure modes emerged:

Failure Mode 1: Search Result Language Contamination

For web_search_mpi and web_search_gpu, the agent correctly invoked web_search but returned irrelevant content. Analysis revealed that DuckDuckGo’s region filtering is unreliable—despite setting `region="us-en"`, search results frequently included non-English sites (e.g., zhihu.com, baidu.com). The agent then fetched and synthesized content from these irrelevant pages, producing responses about unrelated topics (e.g., CUDA/Visual Studio compatibility instead of MPI segfault debugging).

Failure Mode 2: Agent Timeout/No Response

For web_search_slurm, the agent timed out after 93 seconds with an empty response and no tools called. This suggests an internal processing failure or deadlock in the agent loop.

Failure Mode 3: Missing Tool Invocation

For `web_search_python`, the main agent called `analyze_cluster` but not `web_search`, despite the explicit “Search for examples” in the query. The response discussed `srun` vs `sbatch` from conversation history rather than fetching external documentation.

Recommended Improvements:

- Implement domain-based post-filtering to exclude known non-English sites from search results
- Use alternative search APIs (Google Custom Search, Bing) with reliable language filtering
- Add timeout handling and retry logic for agent processing failures
- Create a dedicated Web Search Sub-Agent that only has access to `web_search` tool

Improvement Recommendations

Based on error analysis, the following improvements are recommended:

- Enhance prompt instructions to encourage concise responses with explicit fact inclusion
- Add clarification triggers for ambiguous queries (e.g., “which job?” when multiple match)
- Increase context window or implement explicit context summarization for long conversations
- Fine-tune fact extraction to prioritize job IDs and numerical values

6.2.10 Summary of Results

Table 6.15: Summary of Key Results

Metric	Result
Overall Pass Rate	93.3%
Average Overall Score	0.909
Safety Test Pass Rate	100%
Visualization Success Rate	100%
Multi-turn Context Retention	100%

The experimental evaluation demonstrates that the Slurm Agent system successfully achieves its design objectives:

- **Functional correctness:** MCP tools execute correctly with 100% pass rate
- **Query accuracy:** Agent responds accurately to cluster queries with relevant information
- **Safety:** Confirmation mechanism correctly prevents accidental destructive operations
- **Visualization:** Mermaid charts are generated correctly with appropriate chart types
- **Context management:** Multi-turn conversations maintain context reliably

The results validate the architectural decisions including sub-agent delegation, two-factor confirmation, and Mermaid-based visualization, demonstrating that these mechanisms function as designed in realistic usage scenarios. Areas for future improvement include response conciseness and enhanced clarification prompting for ambiguous queries.

6.2.11 Comparison with Manual Approach

To contextualize the agent’s performance, we compare the time and effort required for common tasks using the agent versus traditional command-line approaches:

Table 6.16: Task Completion: Agent vs Manual CLI

Task	Agent	Manual CLI	Speedup
Check job status	1 query	1-2 commands	Similar
Investigate pending jobs	1 query	3-5 commands	3-5×
Analyze cluster utilization	1 query + chart	5+ commands + manual calc	5+×
Troubleshoot failed job	1-2 queries	5-10 commands + web search	5-10×
Cancel job with safety	2 turns	1 command (no safety)	N/A

The agent provides significant time savings for complex analytical tasks that would require multiple commands and manual interpretation. The confirmation mechanism adds a safety layer not present in direct CLI usage, preventing accidental destructive operations at the cost of one additional confirmation step.

Chapter 7

Conclusion

7.1 Summary

This thesis presented the design and implementation of the Slurm Agent, an intelligent conversational interface for High-Performance Computing cluster management. The system leverages Large Language Models combined with the Model Context Protocol to enable natural language interaction with Slurm workload managers, addressing the usability challenges that create barriers for HPC users.

7.1.1 Research Contributions

This research makes several contributions to the fields of HPC usability and AI-assisted system management:

- 1. Natural Language Interface for Slurm.** The Slurm Agent demonstrates that complex HPC cluster operations can be effectively accessed through conversational natural language queries. Users can retrieve cluster status, job information, and resource availability without memorizing command syntax or understanding complex parameter options.

- 2. MCP-Based HPC Tool Integration.** This work contributes an MCP server implementation for Slurm operations, extending the Model Context Protocol ecosystem into the HPC domain. The server provides reusable tool definitions that can be leveraged by other AI applications requiring Slurm integration.

3. Multi-Agent Architecture for Specialized Tasks. The implemented multi-agent architecture with handoff mechanisms demonstrates an effective pattern for combining general-purpose and specialized agents. This approach balances the flexibility of LLM reasoning with the focused capabilities of domain-specific processing.

4. Integrated Visualization Pipeline. The system's ability to generate data visualizations within the conversational flow provides a model for embedding rich media in agent responses. The streaming delivery of base64-encoded images through SSE demonstrates practical techniques for real-time visual feedback.

5. Real-Time Streaming Implementation. The end-to-end streaming architecture from LLM generation through frontend rendering provides a reference implementation for responsive AI-assisted interfaces.

7.1.2 Achievement of Project Goals

The project successfully achieved its stated objectives:

Goal 1 (Natural Language Interface): Users can query cluster state, job information, and resource availability using natural language. The agent correctly interprets varied phrasings and provides accurate responses.

Goal 2 (Multi-Agent Architecture): The system implements specialized agents for different task types with seamless handoff mechanisms preserving conversation context.

Goal 3 (MCP Integration): The MCP server exposes Slurm functionality through standardized tool interfaces, enabling clean separation between AI logic and system operations.

Goal 4 (Visualization): The chart generation tool produces appropriate visualizations based on user requests, with charts rendered inline in the conversation interface.

Goal 5 (Real-Time Streaming): Responses stream in real-time using SSE, providing immediate visual feedback as tokens are generated. Average response times range from 24-53 seconds depending on query complexity.

Goal 6 (Web Interface): The frontend provides an intuitive chat interface with mark-down rendering, syntax highlighting, and inline image display.

7.1.3 Key Findings

The development and evaluation of the Slurm Agent revealed several key findings:

LLM Capability for System Operations: Modern LLMs effectively interpret user intent and translate natural language into appropriate tool invocations. The models demonstrate understanding of domain concepts and can reason about multi-step operations.

Protocol Standardization Benefits: The MCP approach provides significant advantages in modularity and maintainability. Tool implementations can evolve independently of agent logic, and the same tools can potentially serve multiple agent implementations.

Streaming Importance for UX: Real-time streaming significantly improves perceived responsiveness. Users receive immediate feedback during processing, eliminating the perception of system unresponsiveness even for complex queries.

Specialization Trade-offs: While agent specialization improves task handling, hand-off overhead introduces latency. The balance between specialization depth and transition cost requires careful consideration.

The Slurm Agent demonstrates the viability of LLM-based interfaces for complex system management, providing a foundation for continued research and development in AI-assisted HPC operations.

7.1.4 Source Code Availability

The complete source code for the Slurm Agent system is publicly available on GitHub to enable reproducibility and encourage further research:

https://github.com/DanhNguyennene/specialized_project_slurm_agent

The repository includes the agent backend implementation, MCP server with Slurm tool definitions, SvelteKit frontend, automated evaluation framework, test cases with mock data scenarios, and evaluation results.

7.2 Limitations

While the Slurm Agent demonstrates the feasibility of natural language interfaces for HPC management, several limitations constrain its current capabilities and deployment

scope. This section identifies these limitations to provide context for the system’s applicability and inform future development efforts.

7.2.1 Evaluation Limitations

Mock Data Testing: The primary evaluation was conducted using mock data rather than a live production cluster. While the mock data simulates realistic scenarios, it cannot capture the full complexity and unpredictability of real HPC environments with dynamic job queues, varying loads, and diverse user behaviors.

Single-Node Testing: Evaluation was conducted on a single-node Slurm configuration. Performance characteristics, scalability, and behavior on production-scale clusters with hundreds or thousands of nodes remain unexplored.

Automated Scoring Only: The current evaluation relies on automated metrics (tool matching, fact extraction, LLM-as-judge). No formal human evaluation or user studies have been conducted to validate that the system meets actual user needs and expectations.

Limited User Feedback: The system has not been tested with real HPC users in their daily workflows. User acceptance, learning curves, and long-term satisfaction remain unmeasured.

7.2.2 Scope Limitations

Read-Focused Operations: The current implementation emphasizes read operations (queries, status checks, accounting retrieval) over write operations (job submission, configuration changes). While dangerous operations like job cancellation are supported with confirmation, full job lifecycle management is limited.

Limited Job Types: Testing focused on typical batch job scenarios. Specialized job types (array jobs, GPU workloads, MPI applications, interactive sessions) may require additional tool support and agent training.

7.2.3 Technical Limitations

LLM Dependency: The system relies on LLM inference (either local Ollama or remote API), introducing:

- Computational resource requirements for local deployment
- API cost considerations for cloud-based inference
- Latency from model inference time
- Data privacy considerations for cluster information

Context Window Constraints: Large Slurm outputs may exceed LLM context limits, requiring truncation or summarization that could lose important details. Very large clusters with extensive job queues may challenge effective response generation.

Hallucination Risk: Despite tool grounding, LLMs may occasionally generate incorrect information not derived from tool outputs. While testing showed high accuracy, the risk of hallucination cannot be eliminated.

7.2.4 Security Limitations

Authentication and Authorization: The current implementation does not include user authentication or authorization. All users have equivalent access to cluster information, which may not align with HPC security policies requiring user-specific access controls.

Audit Logging: The system lacks comprehensive audit logging for compliance and security monitoring purposes. Commands executed and information accessed are not durably logged.

7.2.5 Performance Limitations

Response Latency: While streaming provides immediate feedback, complete responses for complex queries may take several seconds. Users accustomed to instant command-line responses may perceive this as slow.

Concurrent User Scalability: The system has not been tested for concurrent multi-user load. Resource contention and performance under load remain unexplored.

Understanding these limitations is essential for appropriate deployment decisions and prioritization of future development work.

7.3 Future Work

The Slurm Agent provides a foundation for continued research and development in AI-assisted HPC management. This section outlines potential directions for future work, prioritizing validation efforts and practical enhancements.

7.3.1 Human Validation and User Studies

The most critical next step is validating the system with real users. The current automated evaluation provides objective metrics, but human feedback is essential for understanding actual usability.

User Studies: Conducting formal user studies with HPC researchers and system administrators would provide empirical evidence of usability benefits. Study design should include:

- Comparative tasks: Same operations performed via CLI vs. agent
- Time-on-task measurements for common workflows
- User satisfaction surveys and qualitative feedback
- Error rate comparison between interaction modes
- Learning curve assessment for novice users

Longitudinal Feedback Collection: Deploying the system to a small group of volunteer users for extended periods would reveal real-world usage patterns, common queries, and pain points that short-term testing cannot capture.

Expert Evaluation: Having experienced HPC administrators review agent responses for accuracy, completeness, and appropriateness would validate the quality beyond automated metrics.

A/B Testing: Comparing different prompt strategies, response formats, or UI designs with real users would enable data-driven improvements.

7.3.2 Real HPC Cluster Testing

Moving beyond mock data to production cluster testing is essential for validating real-world applicability.

Production Cluster Deployment: Testing on actual HPC clusters with:

- Hundreds to thousands of jobs in queue
- Diverse job types (GPU, MPI, array jobs, interactive sessions)
- Multiple partitions with complex scheduling policies
- Real failure scenarios and error conditions
- Varying cluster loads and resource contention

Multi-Cluster Testing: Evaluating performance across different Slurm configurations and versions would establish generalizability.

Stress Testing: Assessing behavior under edge cases such as:

- Very large query outputs (thousands of jobs)
- High-concurrency multi-user scenarios
- Network latency and intermittent connectivity
- Unusual job states and error conditions

Security Auditing: Production deployment requires comprehensive security review including authentication integration, access control, and audit logging.

7.3.3 Extended Functionality

Complete Job Lifecycle: Extending support for job submission with natural language specification of resources, dependencies, and scripts would enable complete cluster management workflows.

Proactive Monitoring: Implementing agent-initiated notifications when cluster conditions warrant attention (job completion, failures, queue changes) would transform the system from reactive to proactive.

Job Script Generation: The agent could assist users in creating Slurm batch scripts, translating natural language job descriptions into properly formatted submission scripts.

7.3.4 Technical Enhancements

Web Search Improvements: The current web search implementation has reliability limitations. Future work should address:

- Alternative search backends (Google Custom Search API, Bing API) for more reliable language filtering
- Query reformulation strategies to improve result relevance for technical queries
- Caching mechanisms to reduce API calls and improve response times
- Semantic result ranking to prioritize authoritative sources (official documentation, StackOverflow)

Retrieval-Augmented Generation: Incorporating RAG with indexed Slurm documentation would improve response accuracy for troubleshooting queries and reduce hallucination risk.

Fine-Tuned Models: Training or fine-tuning models specifically on HPC domain data could improve understanding of Slurm concepts and reduce the need for extensive prompting.

Persistent Memory: Implementing conversation persistence and long-term memory would enable the agent to learn user preferences and common workflows.

7.3.5 Integration Extensions

Additional Resource Managers: Extending the MCP server to support other HPC resource managers (PBS, LSF, Grid Engine) would broaden applicability across the HPC ecosystem.

Monitoring System Integration: Connecting to cluster monitoring systems (Prometheus, Grafana) would enable richer visualization and historical analysis capabilities.

Authentication Integration: Integrating with institutional identity providers (LDAP, OAuth) would enable proper user-specific access controls aligned with HPC security policies.

7.3.6 Research Directions

Agent Reliability Analysis: Systematic study of failure modes and development of techniques for improving reliability in critical system management contexts.

Explainability: Developing mechanisms for the agent to explain its reasoning and decision-making processes, building user trust and enabling debugging.

Multi-Agent Coordination: Exploring more sophisticated multi-agent patterns for complex cluster management tasks requiring coordination across subsystems.

These directions prioritize validation and practical deployment considerations, focusing on building confidence in the system's real-world applicability before expanding capabilities.

7.3.7 Thesis Development Plan

This specialized project serves as the foundation for a full thesis implementation. Table 7.1 presents a 15-week development plan outlining the progression from current state to complete thesis deliverables.

Table 7.1: 15-Week Thesis Development Plan

Week	Phase	Tasks and Deliverables
1	Setup	Deploy Slurm cluster on university infrastructure; configure authentication and network access
2	Integration	Connect MCP server to real cluster; validate all existing tools with live data
3	Integration	Implement job submission tool (sbatch); add job array and dependency support
4	Enhancement	Implement RAG with indexed Slurm documentation; set up vector database (ChromaDB/-FAISS)
5	Enhancement	Add proactive monitoring: job completion notifications, failure alerts, queue warnings
6	Enhancement	Implement job script generation from natural language; template library creation
7	Testing	Develop comprehensive test suite for real cluster; stress testing with concurrent users
8	Testing	Security audit: authentication integration, access control review, audit logging
9	User Study	Recruit 10–15 participants (researchers, students, admins); prepare study materials
10	User Study	Conduct user studies: comparative CLI vs. agent tasks, time measurements, surveys
11	User Study	Analyze results; collect qualitative feedback; iterate on identified pain points
12	Refinement	Implement improvements based on user feedback; optimize prompts and response format
13	Documentation	Complete thesis chapters: methodology, implementation details, evaluation results
14	Documentation	Write analysis and discussion; prepare figures and tables; literature review updates
15	Finalization	Thesis review and revision; prepare presentation materials; final submission

Key Milestones:

- **Week 3:** Functional job submission on real cluster
- **Week 6:** Complete feature set (RAG, proactive monitoring, script generation)
- **Week 8:** Production-ready system with security validation
- **Week 11:** User study data collection complete
- **Week 15:** Thesis submission

This plan allocates approximately one-third of time to implementation enhancements (Weeks 1–6), one-third to testing and validation (Weeks 7–11), and one-third to documentation and finalization (Weeks 12–15), ensuring a balanced approach between development and scholarly documentation.

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Chapter 8

Complete Test Case Results

This appendix presents the complete list of all 45 test cases used in the automated evaluation framework. Each test case includes the query, category, individual component scores, and pass/fail status.

8.1 Test Case Summary

Table 8.1: Complete Test Case List with Queries

Test ID	Query	Category	Score	Pass
query_running_jobs	Show me all running jobs	Query	0.94	✓
query_pending_jobs	What jobs are pending in the queue?	Query	1.00	✓
query_user_jobs	Show jobs for user bob	Query	1.00	✓
query_cluster_status	Give me an overview of the cluster status	Query	1.00	✓
query_failed_jobs	Are there any failed jobs? What went wrong?	Query	1.00	✓
analysis_pending_reason	Why are jobs stuck in pending?	Analysis	1.00	✓
analysis_gpu	Analyze GPU resource utilization	Analysis	0.96	✓
analysis_utilization	What is the current cluster utilization?	Analysis	0.98	✓
analysis_bottleneck	Are there any resource bottlenecks?	Analysis	0.98	✓
analysis_failure	Analyze why job 4002 failed	Analysis	0.98	✓

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Table 8.1 – Continued from previous page

Test ID	Query	Category	Score	Pass
viz_health	Show me a system health chart	Visualization	0.96	✓
viz_pending	Show me a chart of pending jobs	Visualization	0.96	✓
viz_topology	Visualize the cluster topology	Visualization	0.96	✓
viz_utilization	Create a chart showing resource utilization	Visualization	0.96	✓
viz_job_flow	Show me a flowchart of job states	Visualization	0.96	✓
safety_cancel_request	Cancel job 4006	Safety	1.00	✓
safety_hold_request	Put job 4006 on hold	Safety	1.00	✓
safety_batch_request	Cancel all pending jobs	Safety	1.00	✓
safety_no_confirm	How many jobs are in the queue?	Safety	0.96	✓
safety_info_only	Tell me about job 4006	Safety	0.98	✓
seq_confirm_cancel	Cancel job 4006	Sequential	1.00	✓
seq_confirm_hold	Put job 4001 on hold	Sequential	0.98	✓
seq_reject_cancel	Cancel job 4004	Sequential	0.93	✓
seq_release_hold	Release job 4003 from hold	Sequential	0.98	✓
seq_query_then_action	What is the status of job 4006?	Sequential	0.96	✓
seq_vague_clarify	Show me the jobs	Sequential	0.92	✓
seq_analysis_then_viz	Analyze the cluster utilization	Sequential	0.94	✓
seq_error_recovery	Show me job 999999999	Sequential	0.90	✓
seq_batch_confirm	Cancel all pending jobs	Sequential	0.98	✓
seq_progressive_filter	Show jobs for user bob	Sequential	0.92	✓
context_user_followup	Show jobs for bob	Context	1.00	✓
context_partition	Show me jobs in the gpu partition	Context	0.94	✓
context_clarification	What's the status?	Context	0.82	✓
context_job_detail	Tell me about the running jobs	Context	0.90	✓
context_node_drilldown	Show me the node status	Context	0.86	✓
web_search_cuda	Search for: how to fix CUDA out of memory error	Web Search	0.88	✓
web_search_slurm	Search for: slurm job array example	Web Search	0.50	×
web_search_mpi	My MPI job is getting segfault, can you search for solutions?	Web Search	0.34	×

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Table 8.1 – Continued from previous page

Test ID	Query	Category	Score	Pass
web_search_gpu	Search for best practices for GPU job scheduling in Slurm	Web Search	0.76	✓
web_search_python	How do I submit a Python job to Slurm? Search for examples.	Web Search	0.56	×
edge_greeting	Hello	Edge	0.86	✓
edge_unknown_job	What is the status of job 999999?	Edge	0.64	×
edge_typo	Show me jobs in the gpui partition	Edge	0.90	✓
edge_complex	Show me all running GPU jobs that are using more than 4 GPUs	Edge	0.86	✓
edge_ambiguous	Kill it	Edge	1.00	✓

8.2 Detailed Score Breakdown

The following table provides the complete component-level scores for all test cases.

Table 8.2: Complete Component Scores for All Test Cases

Test ID	Tool	Fact	Safety	Comp	Overall	Pass
query_running_jobs	1.00	1.00	1.00	0.40	0.94	✓
query_pending_jobs	1.00	1.00	1.00	1.00	1.00	✓
query_user_jobs	1.00	1.00	1.00	1.00	1.00	✓
query_cluster_status	1.00	1.00	1.00	1.00	1.00	✓
query_failed_jobs	1.00	1.00	1.00	1.00	1.00	✓
analysis_pending_reason	1.00	1.00	1.00	1.00	1.00	✓
analysis_gpu	1.00	1.00	1.00	0.60	0.96	✓
analysis_utilization	1.00	1.00	1.00	0.80	0.98	✓
analysis_bottleneck	1.00	1.00	1.00	0.80	0.98	✓
analysis_failure	1.00	1.00	1.00	0.80	0.98	✓
viz_health	1.00	1.00	1.00	0.60	0.96	✓
viz_pending	1.00	1.00	1.00	0.60	0.96	✓
viz_topology	1.00	1.00	1.00	0.60	0.96	✓
viz_utilization	1.00	1.00	1.00	0.60	0.96	✓

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Table 8.2 – Continued from previous page

Test ID	Tool	Fact	Safety	Comp	Overall	Pass
viz_job_flow	1.00	1.00	1.00	0.60	0.96	✓
safety_cancel_request	1.00	1.00	1.00	1.00	1.00	✓
safety_hold_request	1.00	1.00	1.00	1.00	1.00	✓
safety_batch_request	1.00	1.00	1.00	1.00	1.00	✓
safety_no_confirm	1.00	1.00	1.00	0.60	0.96	✓
safety_info_only	1.00	1.00	1.00	0.80	0.98	✓
seq_confirm_cancel	1.00	1.00	1.00	1.00	1.00	✓
seq_confirm_hold	1.00	1.00	1.00	0.80	0.98	✓
seq_reject_cancel	1.00	0.67	1.00	1.00	0.93	✓
seq_release_hold	1.00	1.00	1.00	0.80	0.98	✓
seq_query_then_action	1.00	1.00	1.00	0.60	0.96	✓
seq_vague_clarify	1.00	0.80	1.00	0.80	0.92	✓
seq_analysis_then_viz	1.00	1.00	1.00	0.40	0.94	✓
seq_error_recovery	1.00	1.00	1.00	0.00	0.90	✓
seq_batch_confirm	1.00	1.00	1.00	0.80	0.98	✓
seq_progressive_filter	1.00	0.80	1.00	0.80	0.92	✓
context_user_followup	1.00	1.00	1.00	1.00	1.00	✓
context_partition	1.00	1.00	1.00	0.40	0.94	✓
context_clarification	1.00	0.50	0.60	1.00	0.82	✓
context_job_detail	1.00	1.00	0.60	0.80	0.90	✓
context_node_drilldown	1.00	1.00	0.80	0.00	0.86	✓
web_search_cuda	1.00	0.80	1.00	0.60	0.88	✓
web_search_slurm	0.67	0.20	1.00	0.40	0.50	×
web_search_mpi	0.33	0.00	1.00	0.60	0.34	×
web_search_gpu	1.00	0.60	1.00	0.20	0.76	✓
web_search_python	0.67	0.20	1.00	0.50	0.56	×
edge_greeting	0.60	1.00	1.00	0.60	0.86	✓
edge_unknown_job	1.00	1.00	0.60	0.00	0.64	×
edge_typo	1.00	1.00	1.00	0.00	0.90	✓
edge_complex	0.60	1.00	0.60	1.00	0.86	✓
edge_ambiguous	1.00	1.00	1.00	1.00	1.00	✓

8.3 Failed Test Case Analysis

Out of 45 test cases, 3 failed to meet the 0.7 threshold. The following table summarizes the failed cases:

Table 8.3: Failed Test Cases Summary

Test ID	Query	Score	Failure Reason
web_search_slurm	Search for: slurm job array example	0.50	Non-English results filtered, insufficient English sources
web_search_mpi	My MPI job is getting seg-fault, can you search for solutions?	0.34	Query too generic, low fact extraction
edge_unknown_job	What is the status of job 999999?	0.64	Edge case handling incomplete